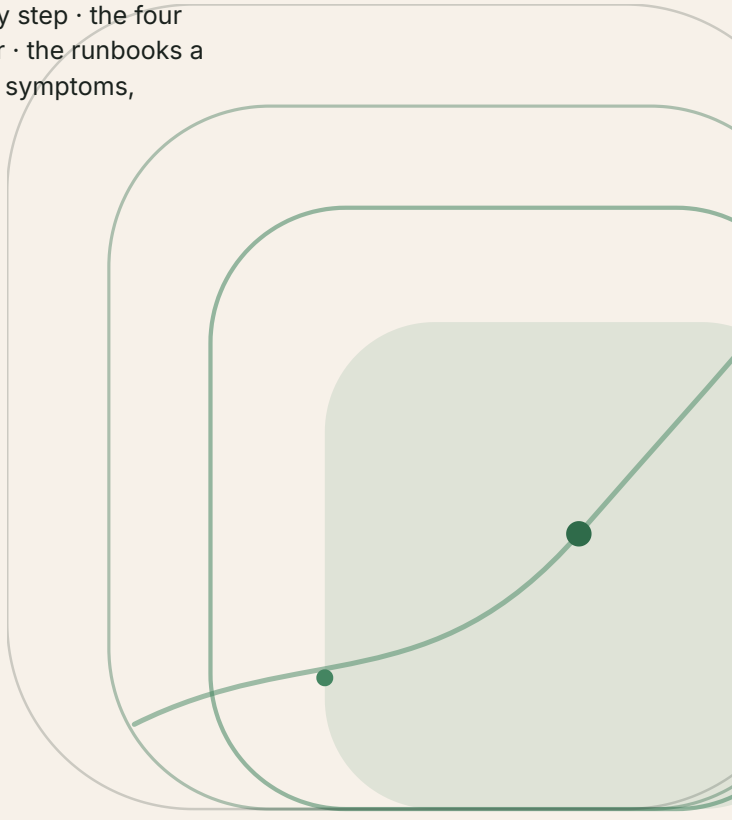
Greecon
SH.P.K

GUIDE 02 · FOR THE PROJECT MANAGER

Project Manager Field Manual

A customer from the first call to day 90: survey, preparing the box, install day, the setup list, commissioning, handover, the morning check and the forms.

FOR	The project manager, for every site
PRINT	Once for the car and the office; print the forms (Appendix A) for each site
COVERS	The job · the rules · the setup list step by step · the four kinds of site · commissioning · handover · the runbooks a project manager uses · forms, glossary, symptoms, shopping lists, contacts



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Greecon Sh.p.k · Durana Tech Park · info@greecon.earth

Contents

The project manager's job, first call to day 90 4
3.1 The rules that never bend 4
3.2 What you work with 5
3.3 The whole path of a customer 6
3.4 Survey visit 7
3.5 Prepare in Greecon (at the office) 8
3.6 Prepare the kit at the office 10
3.7 Install day 12
3.8 The setup list, step by step 13
3.8.1 Order form signed and data terms agreed 13
3.8.2 Prices from the customer's bills 14
3.8.3 Install the box 14
3.8.4 Where each device is on the network 14
3.8.5 Which register holds which reading 16
3.8.6 Every device answers the box 17
3.8.7 Readings arrive and look right 17
3.8.8 Someone is told when the site needs attention 17
3.8.9 Grid meter direction checked 18
3.8.10 Alert limits agreed with the customer 19
3.8.11 Electrical work done by a licensed electrician 19
3.8.12 Commissioned 19
3.9 The four kinds of site 19
A. Hotel or guesthouse (the usual first site) 19
B. Solar plant with grid connection (with an optional battery) 21
C. Farm or greenhouse with irrigation 21
D. Water tank and pump station 22
3.10 After install 22
3.11 Rehearsal: the whole job on a desk 22

Commissioning, handover and what to leave 25
6.1 Commissioning checks by kind of site 25
6.2 Handover 26
6.3 What to leave with the customer 27
6.4 When a check fails 27

Quick runbooks: the ones the project manager uses 29
R0 The routine 29
R1 A customer's box is offline 30
R2 The box is online but a site has no readings, or a device "does not answer" 31
R3 A reading looks wrong 32
R4 A customer did not get a message, or got one that was wrong 32
R5 Too many alerts 34

R6 A person cannot sign in, or someone has left 34
R10 A box is replaced 34
R11 A customer asks for their data, or to leave, or to be deleted 35
R15 Find equipment cannot find a device 36
R16 A device was found at another address 38
R17 The converter's address changed 38
R18 A reading is impossible after the ranges were given 39

Appendices **Forms, words, symptoms, shopping lists and contacts** 40
A Forms to print 40
A.1 Survey form 41
A.2 Install-day checklist 42
A.3 Commissioning record 43
A.4 Handover record 44
A.5 Rehearsal record 45
B Words 46
B.1 Words used in the manual 46
B.2 More words 47
C Symptoms 48
C.1 On site and at the box 48
C.2 A reading looks wrong 50
C.3 A device does not answer the box 50
D Shopping lists 51
H Hotel or guesthouse 51
S Solar plant with grid connection 53
F Farm or greenhouse with irrigation 54
W Water tank and pump station 55
The spares kit and the tools 56
E Contacts and the numbers that never change 57
E.1 Contacts 57
E.2 The numbers that never change 57

The project manager's job, first call to day 90

For: the project manager who surveys, installs, commissions and hands over each site, and then looks after it. **Read with:** the Hardware Reference (PDF 03) (what to buy and how it is connected) and the runbooks (when something is wrong).

You are not expected to be an electrician or a programmer. You are expected to be careful, to write things down, and to say "I do not know yet" instead of guessing. Greecon is built so that nearly everything you do is a form with a plain-words check beside it, and so that the platform tells you when something is not done.

For the equipment in Greecon's kit you type no IP address, no port, no unit number and no register: the kit is made of approved models, their numbers are set at the office, and the box finds them on the network. What you give is what only a person can know: the range printed on a label, who signed a paper, who the electrician was, who is told. Equipment Greecon does not sell (a soil probe, a weather sensor, a battery) is still entered by hand, and section 3.8 says where.

3.1 The rules that never bend

1. **Greecon only reads.** It measures; it never switches anything. Never connect a wire from Greecon's equipment to a pump starter, an inverter's control, a valve or any safety circuit. Nothing in this platform can command equipment, and nothing you install may be made able to.
2. **Anything inside a distribution board is the electrician's work.** Clamping current transformers around live conductors, taking voltage for a meter and connecting a pump's run contact are done by a qualified electrician, under the customer's lock-out procedure, with you watching and writing it down. You do the low-voltage part: the box, the network, the converter, the data cable (you run it to the board; the electrician connects its ends at the meters, because the meters' terminals are inside the board), the labels, and the 12 or 24 volt supply from a plug-in supply. The electrician wires the 4 to 20 mA loops, with the 24 V supply that powers them, as the module's manual draws it.
3. **Never touch the solar panels' wiring (DC), a tank you must climb into, wastewater, or chemicals.** The customer's own staff or installer does that. If the only place for a sensor is there, say it is not possible and write why.
4. **Greecon is not a safety system.** Say so every time, in the words in the sales kit. The protection already in the customer's equipment (float switches, relief valves, trips) stays the safety layer.
5. **Never promise a saving, a percentage or "AI".** Say what Greecon measures and shows. The product shows what the customer's own solar kept on site, at the customer's own prices, from real readings. That is the claim.
6. **Write everything down at the time.** Photographs of every meter's label and display, the wiring before you close the board, the converter's settings, the box's label.

3.2 What you work with

Thing	What it is
Your login	Your own, from the founder. Sign in at `https://app.greecon.earth/login` and choose **Greecon team**. You are asked for a password of your own the first time. Never share it.
Today and **Clients** (the console)	**Today** is where you start: the sites that need a look, the requests waiting and the sites being set up. **Clients** is where you onboard a customer and open their account to set it up. **Requests** lists who has asked for Greecon, and **Team** the people at Greecon who can open a customer's account.
Search	**Ctrl+K** (**⌘K** on a Mac), or **/**, or the **Search** button at the top of the left bar (on a phone, the magnifier at the top right) opens one box. Type a few letters of a client's name to open it, of a request to see who asked, or of a page; inside a customer's account, of a site. **Enter** opens the first result, **Esc** closes.
The setup list	One page per site (`Set up <site name>`) that shows up to twelve steps (a step that does not apply to the site is left out), what is done, what is next, and what to do. Most steps turn **Done** by themselves when the platform sees them work. A few only a person can check, and some of those ask you to write who and when: the papers, who is told, the direction of the grid meter, the limits, the electrician and commissioning.
The install card	One page for each site, linked at the top of the setup list (**The install card**): what to ask the customer and look at before the day, what to bring, what each device is set to and its number on the cable, how it is wired, the converter's settings, the cable rules, what the customer's network is asked for, the visit in order, and three messages to copy to the customer. Print it for the visit.
Approved equipment	The meters and modules in a kit are approved models: Eastron SDM630 and SDM230 meters, a Waveshare Modbus RTU Analog Input 8CH for the 4 to 20 mA transmitters, and a Waveshare Modbus RTU IO 8CH for the pump's run contact. Greecon knows their registers and settings, so nobody types them.
The box	A small computer at the site. It reads the meters and sensors and reports to Greecon. One pasted line installs it. It is also what looks for the equipment on the network when you press **Find equipment**.
The converter	A small unit that puts the meters' cable (RS-485) on the network so that the box can reach them. Greecon sets it at the office.

Thing	What it is
The handover sheet	One printable page per site: where everything is, who is told, who signed the papers, who the electrician was, who commissioned it and when.
The pretend kit	A program for your laptop that pretends to be a whole kit (two meters, the 4 to 20 mA module and the pump contact module), for practice: Rehearsal (section 3.11).

3.3 The whole path of a customer

Stage	What happens	Who	Time
1. First call	Discovery questions (the Sales kit (document 04, not part of these guides)). Decide: worth a visit?	You	20 min
2. Survey visit	Look at the site, fill in the survey form, take photographs, agree what is measured	You (and the electrician if the plan needs one)	1 hour
3. Order form	The customer signs the pilot order form (the Sales kit (document 04, not part of these guides)) and the data terms are agreed; the pilot fee is paid. Nothing goes into the customer's building before both papers are done	Founder or you	
4. Prepare in Greecon	Onboard the customer, make the site from the template, write down that the papers are signed, add the box (section 3.5 below)	You, at the office	30 min
5. Prepare the kit	Install the software on the box at the office, set the unit numbers and the converter, test the kit on the desk, label it, pack it	You	2 hours
6. Install day	Mount, wire, connect, power up; check the labels; press **Find equipment**	You and the electrician	4 to 6 hours

Stage	What happens	Who	Time
7. Commission	Give the ranges, check every number against the real world; confirm the steps only a person can (the electrician's name and licence number among them); commission	You	1 hour
8. Handover	Print the sheet, show the owner the morning message and an alert, sign	You	30 min
9. Look after it	Day 1, 3 and 7 checks, then the morning check (the runbooks)	You	5 min a day
10. Day 60	Decision meeting with the owner, against the three written goals	Founder and you	1 hour
11. Day 90	Convert (order hardware, start the monthly fee) or remove the kit	Founder	

3.4 Survey visit

Bring: this manual, the survey form below, a phone for photographs, a tape measure, a torch, a notebook. No tools are needed to survey; do not open anything.

Ask, in this order (the answers decide what is measured and what it costs):

1. Who decides, and who gets called at night? (Names, mobile numbers, emails: these become **Who is told**. Each alert message carries an *I have seen this* link, so nobody needs a login to acknowledge; the people who should also look at the screens get a login of their own, and a login needs an email address: see 3.8.8.)
2. What went wrong in the last twelve months that cost money or a guest? (Write it in their words. It is the reason they buy.)
3. The last twelve electricity bills (price per kWh, kWh bought each month) and the water bills. (The platform shows what solar kept on site at *their* price; without a price it shows kWh only.)
4. Is there solar? What brand is the inverter and where is its AC output? Is it allowed to send energy to the grid?
5. Where does the supply enter the building? Is it three-phase?
6. Where are the water tank, the pump and its starter? How tall is the tank? What is the pipe's pressure? Has the starter a spare "running" contact? (Without one there is no pump contact module, and "the pump is running but no water is flowing" cannot be raised.)
7. Is there a network socket in the plant room, or cellular signal? (Look at your phone there.)
8. Is there a power socket for the box, and is it on a circuit that stays on?
9. Who is the customer's electrician, and can they be there on install day?

Look at, and photograph:

- The supply entry and the main board (closed): where a meter could be fitted with the clamps around the incoming phases
- The inverter and its AC output cable
- The tank, the pump, the starter, any existing level or pressure gauge and flow meter
- The place for the box: dry, 30 cm free, no heat, within reach of a socket and a network cable
- The route for the data cable from the board to the box (under 100 metres; note any crossing of power cable)

Go or no-go. Go if: there is a way to the internet (cable or cellular), a socket that stays on, a safe place for each meter, an electrician for the board work, and a person who answers when called. **No-go, politely, and write why,** if: the only place for a sensor is unsafe to reach, the customer wants something switched or controlled, or there is no responsible person to receive alerts.

Survey form (one per site; a page in the shared drive):

Field	Answer
Customer, site name and address, date	
Contact: name, mobile, email (first, second, third to be told)	
What went wrong last year (their words)	
Template: Hotel / Solar and grid / Farm / Water station	
Supply: single or three phase; main breaker size (A)	
Inverter brand, size (kW), AC output cable location	
Tank: height (m), water source, pump (kW), starter location, spare running contact on the starter	
Meters needed (grid, solar, load, pump) and where	
Sensors needed (level, pressure, flow, pump running) and where	
Network: customer's cable or cellular; router brand; who can give an address	
Power for the box: socket location	
Electrician: name, mobile, day	
Photographs taken (list)	
Go / no-go, and why	

3.5 Prepare in Greecon (at the office)

You can do all of this before you set foot on site, and it makes the install day shorter.

1. **Sign in** with **Greecon team**. You land on **Today**.

2. **If the customer asked through the web page:** their request is waiting under **Requests waiting** on **Today**, and on the **Requests** page. Press **Onboard** (it opens **Clients** with the form filled in from their request). **Otherwise,** open **Clients** and press **Onboard a client** (top right): company name, owner's name, owner's email. Press **Onboard client**.
3. A box appears: **Client onboarded — save this password now.** That is the owner's temporary password. It is shown **once**. Write it on the handover page, or tell it to the owner by phone. Do not send it by email. The owner signs in at `https://app.greecon.earth/login` with **Client login**, and chooses a password of their own straight away.
   - If a message says someone already has an account with that email, the person is already a customer. Do not use a different address to get round it: add the new site to the customer they already have. (The rehearsal in section 3.11 is the one exception.)
4. In the customer's row on **Clients**, press **Open for setup** (on **Today** the same button is **Open account** for a site that needs a look and **Continue setup** for a site being set up). You land on the customer's **Sites** page. A banner appears at the top: *You are in as Greecon staff, setting it up.* Everything you do from now is recorded under your own name in their audit trail. You can see their people but cannot add people to their account or reset their passwords while inside; their owner does that. **The setup session lasts four hours.** After that the page sends you to the sign-in page: sign in again with **Greecon team** and press **Open for setup** again. What you saved is still there; only what you were in the middle of typing is lost. On install day, open it again when you start commissioning.
5. On **Sites**, press **Add a site** (top right; in the middle of the page while there is no site yet). The form, *New site from a template*, makes the site. Choose **What kind of site**:
   - **Hotel or guesthouse**: water tank and pump, solar, grid. The usual first site.
   - **Solar plant with grid connection**: solar and grid, with an optional battery and load meter.
   - **Farm or greenhouse with irrigation**: soil probe and weather, with the water supply behind it.
   - **Water tank and pump station**: tank, pump and the pump's electricity.
   - **Empty site**: nothing assumed. Use only when none of the above fits.

   The page lists what the template will contain. Enter **Site name** and **Where it is**, check **Currency of the bills** (it starts as ALL) and **The site's own clock** (Europe/Tirane), and click **Create the site and start setting it up**.
6. You land on **Set up** . It says `0 of N steps done`, and highlights what comes next: **Order form signed and data terms agreed** (3.8.1). Do it now, with the signed papers in front of you: write who signed for the customer and on what date, and press **The papers are signed**. At the top of the page is **The install card**: open it and print it. It is your list for the visit (see install day, below).
7. Click **Add the box and get its install command**. A line appears, beginning `curl -fsSL https://api...` ending in a code like `GRC-XXXX-XXXX-XXXX` . **Copy it** (there is a copy button). It is shown once. The code works **once**, for **seven days**. Click **I have copied it**.
   - The code is the box's key. Anyone with the line can install a box as this site until it is used or expires. Treat it like a password: not in email or chat; paste it straight into the box.
   - Lost it, or the seven days passed? On the same step, press **Install a box, or replace this one**. It asks you to confirm, then shows a new line. A new code replaces the old one (the old line no longer works), and the box that uses the new line takes the place of the box now installed, which stops working at that moment.

3.6 Prepare the kit at the office

Do the install of the box's software, and set the kit's numbers and the converter, at the office, on your desk, so that the site visit is only plugging in and checking labels.

The box. Installing the operating system and pasting the install line are still done in a terminal (a keyboard and screen, or SSH); everything after that is done on the setup page.

1. Install the operating system on the box: **Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS 64-bit**, with the SSH server on (you will use it from your laptop on the site's network; nothing can reach it from the internet), one user, a strong password, and the time zone set to Europe/Tirane. (The Hardware Reference (PDF 03) says which box to buy.)
2. Connect the box to your office network and open a terminal on it (a keyboard and screen, or SSH).
3. Paste the install line from section 3.5, item 7. It begins with `curl` and ends with the code. **Paste it exactly.** It runs in seven numbered steps and tells you each one as it goes: checking the box (and that it can reach Greecon and agrees with it about the time), installing Node.js, making the places the software uses, fetching and checking the Greecon software, giving the box its own credential, starting the service, and waiting to hear that Greecon can see the box. It finishes with:

   ```
   Done. Greecon can see this box. On the site's setup page, give each device its address;
   ```

   (That last line is older than **Find equipment**: on the setup page, giving each device its address is now that button, in step 4 of the setup list.)
4. In Greecon, refresh the setup page. **Install the box** now says **Done**, with the box's name, its version and "Heard from N seconds ago". You did not click anything to make it so: the platform saw the box.
5. Label the box with the site name and the date, and pack it.
   - The box keeps its credential when it is moved. Unplug it, carry it to the site, plug it in: it reports from the new place without anything being typed again.

If the line stops with `STOP:` it says what to check and how. If it stopped before step 5, **Giving this box its credential**, fix the cause and run the same line again: the code has not been used yet. From step 5 on the code is used up (it works once), and the same line would stop with "Greecon did not give the software to this box". Then run the line again **with no code at the end**: `curl -fsSL https://api.greecon.earth/edge/install.sh | sudo bash -s --`. That repairs the box and keeps its credential. If it answers "This box has no code and has not been installed before", the box never got its credential: press **Install a box, or replace this one** and use the new line. The commonest stops are: the box cannot reach Greecon (no internet, a firewall that blocks outgoing HTTPS on port 443, or a clock so far out that secure connections fail), or a code already used or older than seven days. A clock that is only minutes out does not stop the install: it prints a WARNING, and the setup page later says "its clock is N minutes fast".

The numbers and the converter

Every kit is made of approved models, and every device has the same number on the cable in every kit. You set them here, at the office. On site you only check the labels.

Device	Make and model	Its number	How it is set
Grid meter	Eastron SDM630	1	On its own display
Solar meter	Eastron SDM630	2	On its own display
Water I/O module (the 4 to 20 mA module)	Waveshare Modbus RTU Analog Input 8CH	3	With the maker's setting tool
Load meter or pump meter, if the site has one	Eastron SDM630 or SDM230	4	On its own display
Pump contact module, if the site has one	Waveshare Modbus RTU IO 8CH	5	With the maker's setting tool
Soil probe (kit F)	The maker's own	6	By the maker's instructions, before it is buried
Weather sensor (kit F)	The maker's own	7	By the maker's instructions, before it is mounted

A battery is the customer's: it is addressed by hand (for example 8). Everything on one cable is set to the same speed: **9600 baud, 8 data bits, no parity, 1 stop bit**.

- **The meters.** Set each meter's number on its own display, one meter at a time (a meter is delivered as unit 1, and two on one cable would answer over each other). The SDM630 is delivered at 9600 baud: check it. The SDM230 is delivered at a speed that depends on its version (the V1.6 manual says 2400): set it to 9600, on the Baud Rate page (the Hardware Reference (PDF 03), section 4.5). Powering a meter on a bench is mains work: the electrician makes up the office's bench lead, and you never wire mains yourself.
- **The two Waveshare modules.** Each is delivered as address 1, and its address is kept in a register: there is no switch. Set it with the maker's setting tool and a USB-to-RS-485 adapter, one module at a time: **3** on the 4 to 20 mA module and **5** on the pump contact module. Use the version of the 4 to 20 mA module that reads current, not the one that reads voltage, and leave its inputs in their default **0 to 20 mA** mode, on purpose: a 4 to 20 mA transmitter then reads 4000 to 20000 microamps, and a wire that has come off reads 0, below the 4 mA of an empty tank, which Greecon can see and say.
- **The soil probe and the weather sensor** (kit F): 6 and 7, set with the USB-to-RS-485 adapter and the maker's instructions, before the probe is buried or the sensor mounted. A second probe takes the next free number.
- **The converter.** In its web page: mode **Modbus TCP to RTU (gateway)** (not "transparent", not "raw TCP", not "Modbus RTU over TCP"), serial 9600, 8 data bits, no parity, 1 stop bit, TCP port 502. Its address on the customer's network is arranged on site, as install day says: a fixed one is best, and Greecon finds the converter either way.
- **Label everything** with its number (the converter with its settings), and photograph the labels.

Test the kit on the desk. The registers of the two Waveshare modules are from the maker's published tables and have not been tried on real hardware, so the first real kit is rehearsed on a desk before it is trusted (section 3.11, step 10). Anything that differs from this manual is written down and reported to Greecon.

3.7 Install day

Before the day. Two or three days before, send the customer the first two messages on **The install card**: the first to their contact, the second to whoever looks after their network. Each is in English, with an *Albanian draft* folded under it: have someone who writes Albanian every day read the draft before you send it (the interface and Greecon's own messages are English only). The day before, check the signed papers, the electrician's number and the kit, packed from the install card's list.

Before anyone touches anything: the signed order form and the data terms are in your hand, the electrician is there, and the customer's lock-out is agreed.

Order of work. The electrician and you work at the same time, in this order, so that nobody waits.

	Electrician	You
1	Walk the site with you; agree where each meter and sensor goes; apply the lock-out	Walk with them; photograph before
2	Fit the grid meter and the solar meter in the board: voltage connections with their fuses as the meter's manual says, the current clamps around the incoming phases, **arrow towards the site**	Mount the box, the converter and the 12 or 24 V supply on a shelf or rail; run the data cable from the converter to the board (the electrician connects it at the meters)
3	Connect the pump's run contact to input 1 of the pump contact module (a dry contact, never the relay outputs); fit the level and pressure transmitters (a wet part: only the electrician or the customer's plumber) and wire their 4 to 20 mA loops with the 24 V supply in series, as the module's manual draws it: level on input 1, pressure on input 2, flow meter on input 3	Check each device's number on its label against the install card (grid meter 1, solar meter 2, water module 3, load or pump meter 4, pump contact module 5) and photograph each label. They were set at the office: do not change them
4	Remove the lock-out only when you both agree the work is closed up	Check the converter's label (gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502); connect it and the box to the network; power up
5	Stay until commissioning is done	Open the customer for setup again and go down the setup list: press **Find equipment**, give the ranges, and write down the electrician's name and licence number

Numbers are set at the office, and only checked here. The numbers and the speeds were set before the kit left (section 3.6). If a label is missing, or two devices of a kind carry the same number, **Find equipment** says **Not sure which**: do not guess, and do not change a number on site. For what to do, see Appendix C and the runbook R15.

Meter settings are read from the meter's own display, never assumed. Different makers, and different versions of the same meter, are delivered with different defaults (the Eastron SDM630 is 9600 baud, and the SDM230 can be 2400). Greecon sets every meter to the same speed, and to its own number, at the office; on site you read them again from the display, and photograph it.

The network, whichever way it comes:

Situation	What you do
The customer's cable and router (best)	Plug the box and the converter into the router. Ask the customer's IT person (or the router's page) to give the converter a **fixed address** that the router will not change (a "DHCP reservation" on the converter's MAC address, or a fixed address outside the router's range). Write it down. You do not type it anywhere: **Find equipment** finds the converter, and finds it again if its address ever changes, but a fixed address does not change under it. The box can take any address.
A fixed address you set yourself	Set the converter's own address in its web page, on the same range as the router.
No usable internet (cellular)	Fit the cellular router from the kit with a SIM, plug the box and the converter into **its** sockets. Greecon supplies the SIM, charged at cost. Check the signal where it sits.
A firewall	The box needs only **outgoing HTTPS (port 443)** to `api.greecon.earth` and, when it is installed, to `nodejs.org`, plus the time service (port 123 UDP, or the box's own). Nothing needs to come in. When someone presses **Find equipment**, the box looks at the site's own network for about a minute: it tries port 502 (or the port the person gives) on the private addresses of that network, such as 192.168.x.x and 10.x.x.x, and only reads; it never writes to anything and never looks outside the site's network. If the customer's network administrator asks, the second message on **The install card** says it in their words.

Before leaving the board: photograph the meter's label (model and serial number), its wiring, its display showing the address and speed, and the converter's settings page.

3.8 The setup list, step by step

Open the site's setup page. Each step is a card. A step opens to show its detail, and stays as you leave it. A step is **Done**, **To do**, **Waiting** for another step, or **Needs attention**; the banner at the top, **Do next: …**, names the step to do next. The platform works most of them out by itself; you only press buttons for what a person must check or say. There are up to twelve steps, always in this order (a step that does not apply to the site is left out): the papers, the prices, the box, finding the equipment, the ranges, the devices answering, the readings, who is told, the direction of the grid meter, the limits, the electrician, and commissioning.

3.8.1 Order form signed and data terms agreed

(the first step: nothing goes into a customer's building without it)

Greecon does not put equipment in a customer's building without a signed order form and agreed data terms (what Greecon reads, who sees it and how long it is kept). Greecon keeps the two papers; the platform cannot see a signature, so you write down that they exist. In the box above the button write **who signed for the customer and on what date**, for example *Signed for the hotel by Besa Krasniqi on 19 October*. The box cannot be left empty. Press **The papers are signed**. The step turns **Done**, *Signed and agreed.*, with your name, the time and what you wrote, and the same goes on the handover sheet. If a paper turns out to be wrong, press **Reopen**: if the site was commissioned, that is taken back too. Do it at the office, before anything else, and bring copies on install day. Commissioning stays grey until this step is done.

3.8.2 Prices from the customer's bills

(optional, but it is what turns kWh into euros)

In this step, open **Enter prices and baseline** (once something is entered it is called **Prices and baseline**; the same form is on the site's page). Enter: currency; the electricity price per kWh from the latest bill; what is paid for energy sent to the grid (blank if nothing); the water price per m³; "a typical month before": the kWh bought and the m³ of water from last year's bills (this is what the platform compares with; **without it no comparison is shown**, which is correct); where those numbers came from; whether the site **cannot send energy to the grid** (tick it if the plant is zero-export); and the time zone (`Europe/Tirane`). **Save.**

3.8.3 Install the box

Done by itself once the box reports (section 3.6). If the box says **Late** or **Offline**, it is not reaching Greecon: see Appendix C.

3.8.4 Where each device is on the network

Nobody types an IP address, a port or a unit number here. Press **Find equipment**. (The button is grey until the box is online and the site has devices; while the box is looking, pressing it again does nothing.)

- The box looks at the site's own network: only private addresses (such as 192.168.x.x and 10.x.x.x), port 502, about a minute at most. It asks each unit number 1 to 10 a few read-only questions, and it never writes to anything. After you press, the page says *The box is looking at the network now.* and follows the box by itself, so you do not reload it. The box picks the request up within about half a minute.
- The platform recognises what answered and matches it to the devices the site is meant to have. It fills in at once only what cannot be a guess: a device with no address yet, found at the unit number the kit gives it, as the model planned. Everything else waits until you press **Use this address**.

Each device gets a line in plain words:

- **Set**: found, and its address is filled in. Where an SDM230 was planned and the meter also answers as a three-phase one, the line adds *It also answers the three-phase meter's total-power register, so it may be a three-phase SDM630: check the model on its label, and if it is one, remove this device and add the three-phase meter from "Add equipment Greecon already knows".*

- **Found**: found, and waiting for you to say yes: at an address other than the one it has, at a unit number other than the kit's, or a meter of the other model than the one planned. Press **Use this address** (or *Use these N addresses*, when there are several) when the line makes sense. The line says where and what, for example *Found at 192.168.1.60, unit 3: a 4 to 20 mA input module, reading 12.0 mA, 9.0 mA, 5.1 mA.* At another number it adds *It is at unit 4, not 1: if it is this site's, use this address, and the install card gives 1 so that the next site looks like this one.* If a single-phase SDM230 is found where a three-phase SDM630 was planned, it adds *It is a single-phase SDM230, not the three-phase SDM630 planned. If you use this address, Greecon reads it as a single-phase SDM230.* The line then keeps saying *Greecon reads it as a single-phase SDM230; the site's plan had a three-phase SDM630.*
- **Not found**: *Nothing that looks like an Eastron meter answers at unit 2. Check the address set on it, the speed, and the A and B wires.* For an optional device the line adds *If the site does not have one, remove it from the list below.*
- **Not sure which**: two of one kind answer at the same number on different converters, or two of a kind answer at numbers that are not the kit's; or something answers but not clearly enough to tell the model (*Something answers at 192.168.1.60, unit 1, but not clearly enough to tell whether it is the three-phase SDM630 planned. Look again; if it is the same, check the speed and the A and B wires.*); or every unit number answers with the same words (*Units 1, 2 at 192.168.1.50 all answered with exactly the same words*), which means the converter is not in Modbus TCP to RTU (gateway) mode. Greecon refuses to guess, because a wrong guess could swap the grid meter and the solar meter, and every figure with it. Read the label on each device to see which is which, then use **Enter the address by hand** under each device, with the number on its label. Numbers are set at the office and are not changed on site: see the runbooks, R15.
- **Enter by hand**: equipment Greecon cannot recognise on a network: the soil probe (6), the weather sensor (7) and a battery. Under the device open **Enter the address by hand**, fill in the **IP address** (the converter's, or the battery's own), the **Port** (502) and the **Unit number**, and press **Save**. (Once a device has an address the link reads **Change the address by hand**.)

When it worked, the step is **Done** (*Every device has an address.*), the panel says *Every device has the address it was found at.*, and the button now reads **Look again**.

Also on this step:

- **A device the site does not have.** An optional one (the pump contact module, the load meter, the pump meter, a battery) shows **The site does not have this: remove it**. Press it and confirm: the device and its readings go. If you removed one by mistake, add it back with **Add equipment Greecon already knows**, at the bottom of the step (the meters and the two modules).
- **The converter is on another part of the network**, or uses another port, or the box is not on your network (a desk rehearsal): open **I know the converter's address**, type the converter's **IP address**, each part without a leading zero (192.168.1.50, not 192.168.001.050), and the **Port** only if it is not 502 (the field can stay empty), and press **Look at this address**. Only private addresses are accepted, and the box does not look at itself (127.0.0.1).
- **Anything else that answered** is listed under *Also answering, and not part of this site's equipment*. Ask what it is before you go on.
- **If the converter's address changes later** (the router gave it a new one) the readings stop. Press **Find equipment** again, then **Use this address**: the runbooks, R17.
- On the box, `greecon-edge find` does the same looking and prints what answers: for a desk, and for support.

- Pressing **Find equipment**, and every change of a device's address or model (old and new), is written in the customer's audit trail under your name.

If it does not work: *No Modbus converter answered on this network. Check that the converter is switched on and plugged into the same router as the box, then find the equipment again. If the converter is on a different part of the network, give its address (I know the converter's address).* Or *A converter answers at 192.168.1.50, but nothing answers behind it. Check each meter's address, the speed on the converter and the meters, and the A and B wires.* Or *The box did not pick this up. Check that it shows as online in the step above, then press Find equipment again.* The button stays grey while the box is not online (*The box has to be online first: it is the one that looks.*) or the site has no devices yet.

3.8.5 Which register holds which reading

- For equipment Greecon knows (the Eastron meters and the pump contact module), nothing is typed: the registers are already filled in. Check them against the maker's manual only if a reading looks wrong.
- For the 4 to 20 mA module (the **Water I/O module**) the page asks, for each transmitter, only for the range printed on its label, in the label's own words:
  - **Level transmitter** (input 1): the **Range on the label** (for a 0 to 5 m transmitter, 5), the **Unit on the label** (metres, or bar for a pressure sensor hung at the bottom of the tank; the page then says what depth it is, for example *0.5 bar holds up 5.0985 metres of water*) and **How tall the tank is inside** (say 4, always in metres): from the bottom to the top of the water when the tank is full. Greecon shows the level as a percentage of the tank, so a full tank reads 100.
  - **Pressure transmitter** (input 2): the **Top of the range** in bar (10 is usual).
  - **Flow meter** (input 3): the **Top of the range** in litres a minute (the form says lpm).
  - A comma works as a decimal point (2,5 is 2.5).
  - A transmitter the site does not have (no flow meter): tick **Not fitted at this site**. When you save, a box asks *Take away pump flow from this site? The reading and its history are removed. This cannot be undone.* Say yes only if the site really has no such transmitter; the reading is then gone, so that nothing waits for it.

  Press **Save the ranges**. It says *Saved. The box picks it up within a minute.*, and under each transmitter the form says what it set, for example *Set now: 4 mA is 0 % and 20 mA is 125 %.* (a 0 to 5 m transmitter in a tank 4 m tall: 5 ÷ 4 × 100 = 125, so that a full tank, at 16.8 mA, reads 100). A transmitter shorter than the tank is refused, in words: *The transmitter reads up to 3 m and the tank is 4 m tall: it cannot see the top of the water. Fit a transmitter with a longer range.* To change a range, give it again; the form does not show the level's answer back, so type both of its numbers.

- For equipment Greecon does not know (the soil probe, the weather sensor, a battery): one form per reading, from the maker's register table. Under the reading's name (for example **Soil moisture**) press **Choose the register**, and fill:
  - **Register kind**: **Input** for a manual's 3xxxx, **Holding** for 4xxxx (the form starts on Holding).
  - **Address**: counted from zero, so a manual's **30053 is address 52**, and 40001 is address 0.
  - **Kind of number** and, for two-register numbers, **which half comes first** (the usual is the high half first).
  - **How it becomes the reading**: **Multiply and add** (a meter that reports watts for a reading in kW: multiply by 0.001), or **A straight line, such as 4-20 mA**, where **Signal at the bottom / top** are what the register holds at the two ends of the sensor's signal and **Reading at the bottom / top** are the reading's own values at those two ends (the maker's table says what).
  - **Which part of the value**: "Only when positive" and "Only when negative, as a positive number" split a meter that reports import as positive and export as negative into two readings (the templates do this for you).
  - **Save the register.**
- **Test any register before relying on it**, from the box:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
float32
greecon-edge modbus-test --host 192.168.1.50 --unit 5 --type discrete --address 0
```

  The first reads a meter, the second the pump contact module (`--type discrete` reads a digital input). Each prints the raw numbers and the value. If it prints `Could not read`, the problem is the wiring or the settings, not Greecon.
- A reading the site does not have holds the list at this step: for a transmitter, tick **Not fitted at this site**; for an optional device, press **The site does not have this: remove it** in step 4 of the setup list.

3.8.6 Every device answers the box

Done by itself. Each device shows **Done** ("Answering (N readings)") or **Needs attention** with the reason in words ("no answer from 192.168.1.50:502 (unit 2)"); on the box, `greecon-edge status` lists each device as **ok** or **NOT OK**. Look in this order: the converter's address (the router may have given it a new one: press **Find equipment** again, then **Use this address**), the number on the device's label against the install card, the **speed** at the converter, **A and B swapped** on the data cable, the converter not in gateway mode.

3.8.7 Readings arrive and look right

Done by itself when each reading has arrived, in a believable range. A reading that cannot be real says so, for example *-31.2 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.* On a transmitter that is nearly always a wire that has come off (the module then reads 0, below the 4 mA of an empty tank, so the reading comes out below zero), or a range or a tank height that was given wrongly: the runbooks, R18. A tank level may read up to 5 points over 100 or under 0 without being called impossible. **Pump running** reads 1 while the pump runs and 0 while it is stopped.

3.8.8 Someone is told when the site needs attention

Open **Who is told** (the step has the link):

- **Add a person** for each name from the survey, **in the order they should be told.** Each needs an email, a mobile number **with its country code** (`+355 69 123 4567`), or both. The first person hears first.
- **Save the list.** Then, beside each person, **Send a test message.** It says what happened to the email and to the text ("Accepted for delivery to +355 ••• ••• 567. Ask Besa whether it arrived."). Tests are limited for the whole account: six an hour (a person with both an email and a number uses two of the twelve test messages allowed). If you reach the limit the page says "Test messages are limited to 6 an hour. Try again later."
- **Ask each person** whether it arrived: in the inbox, not in spam; on the phone.
- **Tell each person on the list how to say they have seen an alert.** Every alert message has an *I have seen this* link: one tap on the phone, then one button, no sign-in, good for 72 hours; it stops the chain of messages. Pressing **Acknowledge** on the **Alerts** page does the same, for a person who can sign in. Replying to an email or a text does not. So nobody needs a login for this; give a login of their own (the Operator role is enough) to the people who should also look at the screens: open **Settings**, then **People**, and look at who has a sign-in. If someone who should have one has none, the customer's owner adds them under **Add a person** with **Add person** (a Greecon support session can see the people but cannot add them), and the new person is given a temporary password, shown once. An Owner or an Administrator can press **Acknowledge** too; a Viewer or an Auditor cannot (the link in a message works for anyone it was sent to).
- **Give everyone an email address as well as a number.** The note that a colleague has acknowledged an alert, and the note that it has cleared, go by email only. A person who has only a phone number is texted the urgent alert and then hears nothing more: they may still set off for something that is already being dealt with.
- Back on the setup page, write what you saw if you like, and press **I checked this.** It is greyed out until the list has someone on it and a test has been sent, because a site nobody is told about must not be commissioned.

How alerts reach people, for the owner (say this at handover, in these words): **Act now** goes by email at once and by text ten minutes later if nobody has acknowledged it; thirty minutes after the alert began, the next person is told by email and text, and so on down the list. Saying you have seen it means pressing the *I have seen this* link in the message (no sign-in, good for 72 hours) or **Acknowledge** on the **Alerts** page (for a person with a login). **Check today** is one email, held until morning if it is night. Beyond those, three short notes only: one to the people who were told when somebody **acknowledges** the alert (who took it), one when the problem has stayed clear for 15 minutes (**Resolved**), and one **morning message** at 07:30 saying whether everything is all right. Quiet hours (22:00 to 07:00) hold everything except Act now.

3.8.9 Grid meter direction checked

(only on sites with a grid meter) Greecon cannot see which way the meter's clamps point. A reversed clamp swaps what the site buys with what it sells, and every figure with it. **With the sun down or the solar switched off, switch on a known load** (a kettle or a heater). **Grid import must go up.** If **Grid export** goes up, the clamps are the wrong way round: the electrician turns them (with the lock-out on), you check again. Then press **I checked this**, with a note of what you saw. If a clamp is left reversed, Greecon notices on its own at night or in cloud (energy cannot go out of a plant that nothing is producing for): the energy picture says so, and after five minutes a **Check today** alert, *The grid meter may be fitted the wrong way round*, opens. At midday it cannot tell, which is why this check stays.

3.8.10 Alert limits agreed with the customer

The step lists each reading that has limits, with what applies. The standard ones: tank **Critical below 15 percent** (an Act now alert), **Low below 35 percent** (a Check today alert); pump **No flow at or below 0.2 litres a minute** (an Act now alert once it has held for 30 seconds, raised only when the box reads both a flow and the pump's running state); battery **Low below 25 percent**. Irrigation advice, **frost** and solar surplus have no standard value, and the platform says nothing about them rather than guess. (Frost: an **Act now** alert opens when the air at the field reaches the grower's own **Frost alert at or below** value and stays there for two minutes. Ask the grower at what temperature the crop is in danger and in which weeks; enter it on the weather sensor's air temperature.) **Go through them with the customer.** To change one, open the **Standard** drop-down beside the reading in this step (it says **1 set**, **2 set** once changed), type the numbers, press **Save limits**, and when they are agreed press **I checked this.** The same drop-down is in the **Limits** column under **Readings** on the site's **Equipment and readings** page. If the pump's running contact reads the wrong way round, the no-flow alert opens while the pump is stopped: the "Pump running" check in section 6.1 finds that. Agree numbers the customer would act on: an alert nobody acts on trains people to ignore the next.

3.8.11 Electrical work done by a licensed electrician

Fitting meters in a board and wiring transmitters and contacts is the licensed electrician's work, under the customer's lock-out. It is not yours. When the electrician has finished and the lock-out is removed, ask for their name and licence number, photograph the wiring, the labels and the displays, and check that the lock-out has been removed. Then write in the box the electrician's **name and licence number**, for example *Arben Hoxha, licence E-1234*, or that no electrical work was needed. The box cannot be left empty. Press **The electrical work is done**. The step turns **Done**, *Done by a licensed electrician.*, with your name, the time and what you wrote, and the same goes on the handover sheet. The platform does not check the licence: what you write is the record. Do not press it while the work is open or the lock-out is still on. Commissioning stays grey until it is done.

3.8.12 Commissioned

Greyed out until everything above is done (the prices are optional and do not hold it; the papers and the electrician's name and licence number do). Then it says: *On site, compare what Greecon shows with what you can see: the meter's own display, the tank, the pressure gauge. When they agree, confirm here.* Do the checks for the site's kind in section 6.1. Press **Commission this site**, with a note. The progress line then reads **Commissioned.**, and your name and the date are on the handover sheet.

3.9 The four kinds of site

A. Hotel or guesthouse (the usual first site)

What it is: water tank and pump, solar on the roof, and the grid. **What is measured:** grid import and export (one three-phase meter at the supply entry), solar production (one three-phase meter on the inverter's AC output), tank level, line pressure, pump flow, pump running.

Bring: two Eastron SDM630 meters with current clamps, an RS-485 to Ethernet converter, a Waveshare 4 to 20 mA module (Modbus RTU Analog Input 8CH) and a Waveshare pump contact module (Modbus RTU IO 8CH) with their 24 V supply, a level transmitter and a pressure transmitter (and a flow meter if the owner

wants dry-run protection), the box, a small UPS, cable, DIN rail and enclosure (the Hardware Reference (PDF 03)). The numbers of the meters and the modules were set at the office. **Ask the customer for:** twelve electricity bills, the water bills, the people to tell.

Wiring in one picture:

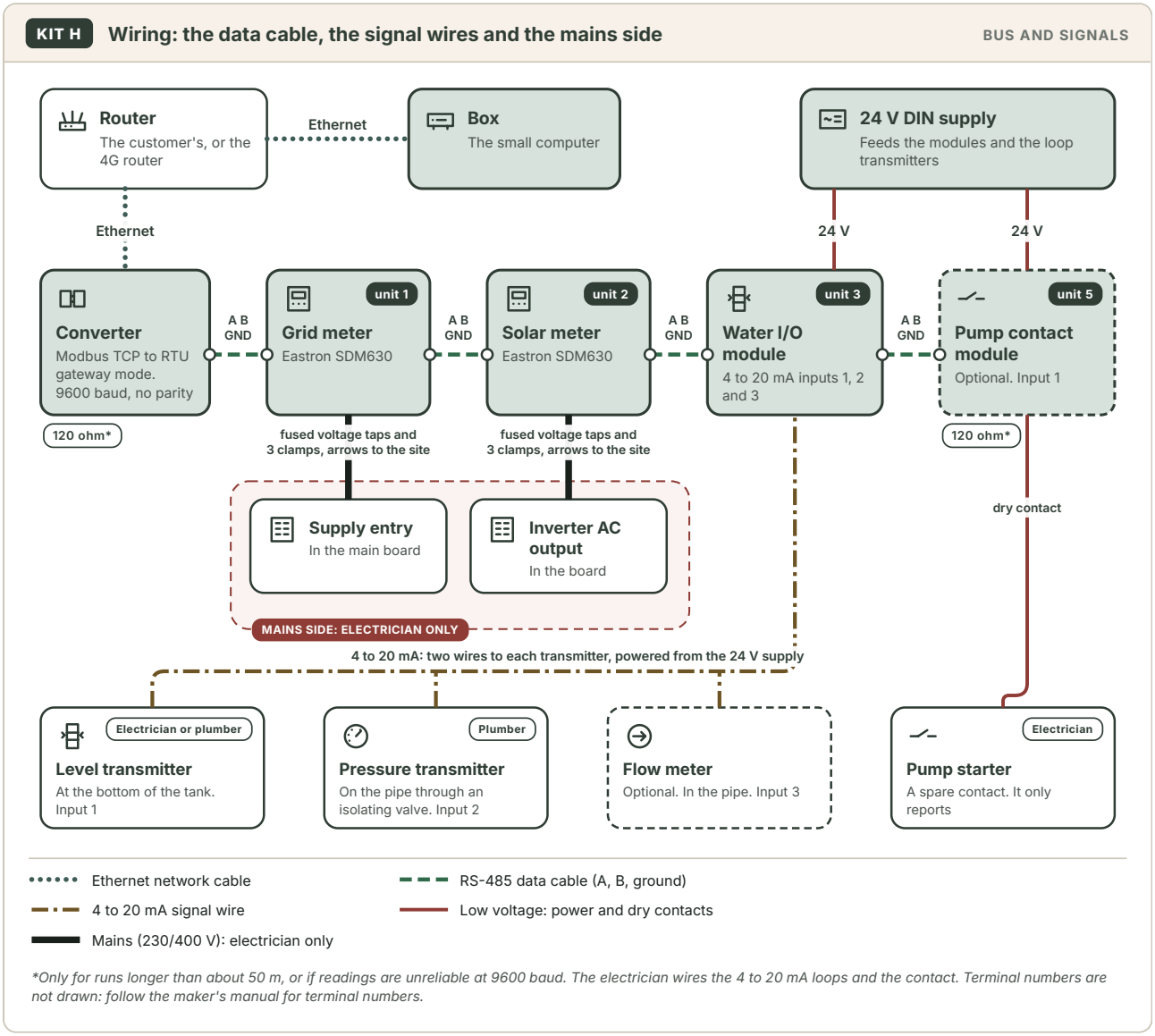


Figure H-2. Kit H. Wiring: one RS-485 cable from the converter through the grid meter, the solar meter and the water module to the pump contact module, with the 4 to 20 mA loops into the water module and the pump starter's contact into the pump contact module. Mains work is inside the dashed box.

Set up in Greecon: template **Hotel or guesthouse** (four devices: the grid meter, the solar meter, the water I/O module and the pump contact module, which is optional). Press **Find equipment**; the registers of all four are known. Then give the ranges (3.8.5): the level transmitter's range and the tank's height, the pressure transmitter's range and, if there is one, the flow meter's. The pump's running state comes from the pump contact module. Prices (3.8.2) so that "your solar kept N kWh on site, worth about X at your price of Y" can be said.

Agree with the owner: the tank limits (the standard 15 and 35 percent are a start: a tank that feeds a fifty-room hotel on a hot day should alert earlier), the no-flow level if the pump normally runs at a trickle (the standard is 0.2 litres a minute), who is told for water and who for electricity, and what "Act now" at 02:00 means for them.

Typical problems: the solar meter reading zero at midday (a closed clamp on the wrong cable, or the inverter is zero-export limited); the level reading wrong (a vented cable blocked, the transmitter not at the tank's bottom, or the range or the tank's height given wrongly); the pump-running contact inverted (reads running when stopped), which also opens the no-flow alert, an Act now, while the pump is stopped (3.8.10).

B. Solar plant with grid connection (with an optional battery)

What is measured: solar production (inverter AC output), grid import and export, optionally the site's own consumption (a load meter) and a battery's charge and power. It works **with any inverter brand**, because it measures at the AC cable with its own meter and not through the inverter maker's portal.

Bring: two or three SDM630 meters (grid, solar, optional load), converter, box, UPS. For a battery: the maker's Modbus register table (without it the battery is left out; do not guess registers). The numbers of the meters were set at the office; a battery is addressed by hand.

Ask the customer: the electricity price, **what is paid for energy sent to the grid, if anything**, and whether the plant is **allowed to send energy to the grid** (if not, tick *This site cannot send energy to the grid* in prices). Twelve bills.

Set up: template **Solar plant with grid connection**. Press **Find equipment**, and take off the battery and the load meter if they are not fitted (3.8.4). To take one off, press **The site does not have this: remove it**. A battery is not recognised on a network: its address and its two readings are entered by hand, from the maker's table. Do the direction check with care: on a plant that exports at noon, do it at night or with the inverter off.

Commissioning: at midday, solar production in Greecon agrees with the inverter's own display within about 5 percent, and grid export is the difference between production and what the site is using.

C. Farm or greenhouse with irrigation

What is measured: soil moisture (per zone) and air temperature and humidity at the field, and the water supply behind it (tank, pressure, flow, pump).

Bring: a soil probe and a weather sensor that speak Modbus RS-485 and that document their registers (without documented registers the sensor cannot be mapped), the 4 to 20 mA module for the tank and pressure and the pump contact module if there is a pump to watch, the box, a cellular router if there is no network in the pump house. The probe's number (6) and the sensor's (7) are set before the probe is buried and the sensor mounted. The box cannot recognise a probe or a sensor, so their addresses and their readings are entered by hand (steps 4 and 5 of the setup list), from the makers' manuals; the water side is as in the hotel.

Ask the grower: which crop, which soil, how deep the roots are, where in the field is typical (not the wettest or driest spot). **Greecon gives no irrigation advice until the grower's own "start irrigating below" value is entered,** because that number depends on the soil, the crop and the probe's depth and nobody can set it for them. After the first week of readings, ask the grower what reading means "time to water" and enter it as the limit. **Ask, too, at what air temperature frost damages the crop and in which weeks of the year.** Enter it as the **Frost alert at or below** limit on the air temperature (the number depends on the crop and how far on it is, so Greecon has none of its own, and says nothing about frost until it is set). A frost alert is an **Act now** message in the night: agree who is woken. **Map the air temperature as a signed number** (int16): below zero, an unsigned register reads a frosty morning as a hot afternoon.

Write in the handover: the probe's depth and where it is. Soil readings are only as good as the contact between probe and soil.

D. Water tank and pump station

What is measured: the tank's level, the line pressure, the pump's flow, whether it runs, and optionally the pump's electricity use (a single-phase meter such as the SDM230).

Ask: what the pump does when the tank is low and when it is full, and who is told today. The pipe's pressure range decides the pressure transmitter's range.

Set up: template **Water tank and pump station** (three devices: the water I/O module, the pump contact module and the pump meter, the last two optional). Press **Find equipment**, give the ranges, and take off what the site does not have (the pump meter, the pump contact module) with **The site does not have this: remove it**. This is the simplest site and a good first rehearsal.

3.10 After install

When	Do	Look for
Day 1	Open **Today**; phone the owner: did the morning message arrive?	Box online; readings current; the message reached them
Day 3	Same; look at one alert if one happened	No alert nobody understood
Day 7	Sit with the owner for 15 minutes: **was every alert useful? what would they change?** Adjust limits.	Alert count at most one a day; every one with a clear action
Every morning	The morning check email and **Today** (the runbooks); each live site shows *Day N*	A silent box, or a site that needs a look
Day 30	Short call; ask what they have done because of Greecon	One thing, in their words, for the sales story
Day 60	The decision meeting, against the three written goals: data current at least 95 percent of the time; the owner says they read the morning message on at least 5 days in 7 (Greecon does not track opens: ask, and write the answer down); at least one real event caught	Convert, extend by up to 30 days once, or remove
Day 90	Convert (order hardware at the agreed price; the monthly fee starts at day 91) or take the kit away within 14 days	Data exported to the customer either way

3.11 Rehearsal: the whole job on a desk

Do this once before the first real site, once more with the first real kit, and again when anything in this manual changes. It takes an afternoon and needs no hardware except the box (or a spare computer running Debian, Ubuntu or Raspberry Pi OS) and your laptop.

1. **On your laptop** (Node.js 22 or newer): get the repository, then run:

   ```
   node scripts/pretend-meter.mjs --kit --port 5020
   ```

   It listens for any machine on the network and answers as a whole hotel kit: unit 1 the grid meter, unit 2 the solar meter, unit 3 the 4 to 20 mA module (the tank at 12 mA, the pressure at 8.96 mA, the flow at 5.12 mA) and unit 5 the pump contact module, with its input 1 closed. Leave it running: it prints each question it is asked, so that you can watch the box look.

2. In Greecon, onboard a client named **Rehearsal**, open it for setup, and make a site from the **Hotel or guesthouse** template. As the owner use **a second address of your own** (a plus-address such as `you+rehearsal@yourmail.example`, if your mail allows it): your own team address already has an account, an address can have only one, and Greecon refuses it. This is the one case where a different address is right.

3. On the setup list write that the papers are signed (for the rehearsal, *Rehearsal: signed by me today*) and press **The papers are signed**. Add the box, copy the line, and paste it on the box (section 3.6). Watch **Install the box** turn **Done**.

4. **Find the kit.** The kit listens on port 5020, not 502, so under **Where each device is on the network** open **I know the converter's address**, type **your laptop's address on the network** (the one the box can reach, such as 192.168.1.60; not 127.0.0.1, which the box takes for itself) and the port **5020**, and press **Look at this address**. After a minute at most the grid meter, the solar meter, the water module and the pump contact module are each **Set**, with lines such as *Already set, and answering at 192.168.1.60, unit 1: an Eastron meter, reading 1.80 kW at 231 V.* (your laptop's address in place of 192.168.1.60). The solar meter reads 5.20 kW; the water module is *a 4 to 20 mA input module, reading 12.0 mA, 9.0 mA, 5.1 mA*; the pump contact module is *a digital input module, reading input 1 is closed*. The step is **Done**. You typed no unit number and no register.

5. **Give the ranges.** Under **Which register holds which reading** type **Level transmitter** range 5 and tank 4, **Pressure transmitter** 10, **Flow meter** 200, and press **Save the ranges**. Within a minute the readings arrive: **Tank level 62.5 %**, **Line pressure 3.1 bar**, **Pump flow 14** litres a minute and **Pump running 1**, with **Grid import** 1.8 kW and **Solar production** 5.2 kW. They come from the kit's signals and your ranges: the kit's 12 mA is the middle of the level transmitter's range, 2.5 m of 5 m, which in a tank 4 m tall is 62.5 percent of the tank ((12 − 4) ÷ 16 × 5 ÷ 4 × 100); 8.96 mA is 3.1 of 10 bar ((8.96 − 4) ÷ 16 × 10); 5.12 mA is 14 of 200 litres a minute ((5.12 − 4) ÷ 16 × 200). If a number differs, look at the ranges you typed.

6. Watch every step turn **Done**. Then add yourself under **Who is told**, send a test message, write for the electrician that no electrical work was needed (the box cannot be left empty), and confirm the steps that need a person. **Commission** the rehearsal site.

7. Stop the pretend kit. After about five minutes **Today** (under **Needs a look**) says the readings are out of date and the box says a device does not answer. Start it again. Then rehearse an alert: in the limits step open the tank's limits, give **Low below** 80 and **Critical below** 70 (both above the tank's 62.5 percent), and press **Save limits**: **Tank level is critical** is an **Act now**. It arrives by email at once and, if you added your number, by text ten minutes later unless you **Acknowledge** it on the **Alerts** page, which then tells the people who were told who took it.

8. Print the handover sheet. Download the readings from **Reports** (**Downloads**). Open the file in a spreadsheet.

9. When finished, remove the rehearsal site's equipment and then the site (on the site's **Equipment and readings** page: **Remove** beside each piece of equipment, then **The site itself**: a site that still has equipment cannot be deleted), or suspend the client from **Clients** (**More**, then **Suspend this client**, and confirm), so that it does not appear in the morning check. If you will use the Rehearsal client to show **Who is told** and **Reports** in a sales demonstration (the Sales kit (document 04, not part of these guides), section 4), keep it and its site until then, and start the pretend kit before the meeting: with it stopped the site shows as out of date.
10. **With the first real kit,** do steps 1 to 6 again with the real converter and devices on your desk in place of the pretend kit, and write down anything that differs from this manual (a number, a speed, a reading, a sentence on the screen). The registers of the two Waveshare modules have only been tried against a pretend module.
11. **To practise the by-hand form** (equipment Greecon does not sell), use the older pretend meter, which is still there:

    ```
    node scripts/pretend-meter.mjs --units 6,7 --port 5021
    node scripts/pretend-meter.mjs --units 6,7 --port 5021 --tank 12 --watts -1800
    ```

    It answers as units 6 and 7 with plain registers: a power at input registers 52 and 12, and at registers 0 to 3 the tank level in tenths of a percent, the pressure in hundredths of a bar, the flow in tenths of a litre a minute and 1 or 0 for the pump running. The second line makes the tank read 12 percent and a plant export. On a second rehearsal site from the **Farm or greenhouse with irrigation** template, give the **Soil probe** your laptop's address, port 5021 and unit 6 with **Enter the address by hand**, and map **Soil moisture** with **Choose the register**: **Input**, address 0, **Multiply and add**, multiply by 0.1. It reads 62 percent with the first line.

If a step in this manual does not match what you see, fix the manual (it is a file in `docs/launch/`), or tell the founder. It is meant to be corrected by the person who uses it.

Commissioning, handover and what to leave

Commissioning is checking, once, on site, that what Greecon shows is what the real world is. Handover is the thirty minutes after it: the owner sees the product working on their own equipment, and is left with what they need. Do them in this order, on install day or the day after, with the electrician still there for the checks that need them. The checks are the ones for the site's kind; the four site packs (PDFs 04 to 07) print them as a record with room for the two numbers, and Appendix A has a blank record for any site.

PROJECT MANAGER ELECTRICIAN CUSTOMER

STOP

Nobody starts, stops or opens equipment for a check except the owner's own person. They know what else it affects. The project manager reads and compares; the electrician works on the mains side under the lock-out; Greecon only reads.

6.1 Commissioning checks by kind of site

Compare, in this order, and write the pair of numbers on the handover.

Check	How	Pass
Every meter's display against Greecon	Read the meter's display (kW or W) and Greecon at the same time	Within 5 percent, or 1 kW on a small load
Grid direction	A known load on, the sun down: grid import rises	Import up, export unchanged
Solar at night	Solar production	Zero, or a few watts
Solar at midday	Against the inverter's display	Within about 5 percent
Tank level	A dipstick or the sight gauge at two levels	Within 5 percentage points
Pressure	Against the pressure gauge on the same line	Within 0.2 bar
Flow	A bucket test (fill a measured volume, time it), or the flow meter's own display	Within 10 percent
Pump running	The owner starts the pump; Greecon shows running within 30 seconds	Changes both ways
Soil probe	In the air (low), in a glass of water (high), in the soil (plausible)	Three different readings

Check	How	Pass
4 to 20 mA channels	With a loop calibrator in place of the transmitter, send 4, 12 and 20 mA	The reading at the bottom, the middle and the top of the transmitter's range (for a 0 to 10 bar transmitter: 0, 5 and 10 bar)
A test message	Each person on the list	Arrived
Power cut	Unplug the box for 30 seconds, plug it in	Back reporting within 3 minutes, and the box says nothing about readings lost
Internet cut *(the first install, and any time the box or its software is new)*	Cut **only the site's internet** for ten minutes: unplug the router's uplink cable (the one that goes to the wall or the modem), or take the SIM out of the cellular router. **Leave the box and the converter plugged into the router**: if the box cannot reach the converter it reads nothing, and there is nothing to test. Then restore it	`greecon-edge status` shows a queue while it is out, and a few minutes after it returns the queue is empty and the readings from the gap are in the downloads. **Write the result down: it is the evidence for the statement that the box keeps its readings.**

On the tank's channel the top of the range reads more than 100 when the transmitter is longer than the tank (a 0 to 5 m transmitter in a tank 4 m tall reads 125 at 20 mA), and Greecon may call that impossible while you test: that is the test, not a fault. A full tank reads 100 (16.8 mA in that example). The module itself must read 4000, 12000 and 20000 at 4, 12 and 20 mA:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 0
```

(The address is 0 for input 1, 1 for input 2, 2 for input 3.) Put the transmitter back when you finish.

Do not **start, stop or open** any equipment yourself. The owner's own person does, because they know what else it affects.

6.2 Handover

1. On the setup page open **Open the handover sheet**. Check it, and print it (**Print or save as PDF**; the printed page leaves out the menus). It names the customer and site, says where each device is (address, unit number, registers), who is told and how to reach them, what was commissioned and by whom, who signed the papers and who the electrician was, and how to reach Greecon (`info@greecon.earth`).
2. Show the owner, in ten minutes, in this order: the **Overview** (is everything all right, in one sentence); **Energy now**, the flow moving from the solar to the site and the grid; **Value**, what the solar kept on site this month (and the prices it uses); **Who is told** (under **Settings**) and a live **test message** arriving on their phone (and that each message carries an *I have seen this* link, so that nobody needs a login to acknowledge an alert); **Reports**, under **Downloads**, where **their data is theirs**: every reading for the last 31 days, hourly averages for up to a year, the alerts, the audit log and how the account is set up can be downloaded whenever they want (and if they stop, they can ask Greecon for every reading it holds, as a file).
3. Say: *Greecon watches your equipment and tells you. The protection built into your equipment remains the safety layer. You will get one message each morning. If there is nothing to do, it says so.*

4. Leave: the printed sheet (signed), the box's label, a card with `info@greecon.earth` and your phone.
5. Photographs and the survey form go in the shared drive, in a folder named for the customer.
6. The same day, send the customer's contact the third message on **The install card** (its subject is *Greecon is now watching* and the site's name), with the handover sheet attached. It is in English, with an *Albanian draft* folded under it: someone who writes Albanian every day reads the draft before it is sent.
7. Press **Back to the Greecon console** (the banner's button) to leave the customer's account: it returns you to **Today**.

6.3 What to leave with the customer

Leave the customer with these, and nothing more:

- ☐ **The printed handover sheet, signed.** It says where each device is, who is told, what was commissioned and by whom, and how to reach Greecon.
- ☐ **The box's label** on the box: the site name and the date. Never a code or a password written on the box.
- ☐ **A card** with `info@greecon.earth` and your phone number.
- ☐ **Their own logins.** The owner chose a password of their own at their first sign-in; the people who should also look at the screens have a login of their own (the Operator role is enough), and everyone on the **Who is told** list knows about the *I have seen this* link in each alert message, which needs no login. The temporary passwords were told by phone, never by email.
- ☐ **The words** in section 6.2: what Greecon is and is not, how an alert reaches people, and that their data is theirs.

Do not leave: the install line or its code (it is the box's key, and it has been used), any password written down, or a promise of a saving.

What you keep, for Greecon: the survey form and every photograph (every meter's label, the wiring before the board was closed, the converter's settings, the box's label) in the shared drive, in a folder named for the customer; the commissioning record with its pairs of numbers, and the internet-cut result, which is the evidence for the statement that the box keeps its readings; and a line in the hardware register for each part: the make and model, the supplier, the date, the serial number or the batch, and the invoice.

6.4 When a check fails

A check that fails is not commissioned. Do not press **Commission this site** until it passes. Look, in this order, at what the table says; the longer answers are in Appendix C and in the runbook for a reading that looks wrong.

Check that fails	Look at first
A meter's display and Greecon differ by more than 5 percent	The model fitted against the one on the install card (an SDM630 fitted where an SDM230 was planned is read as an SDM230 and shows one phase's power only); on a meter with 5 A current transformers, the CT ratio set in the meter's menu; a clamp on the wrong phase or cable; the two meters' numbers swapped (the meter at the supply entry must be number 1). Test the register from the box with `greecon-edge modbus-test` against the meter's display

Check that fails	Look at first
Grid direction: export rises when the load is switched on	The clamps are the wrong way round: the electrician turns them, with the lock-out on, and you repeat the check
Solar reads more than a few watts at night, or nothing at midday	The solar meter's clamps are on the wrong cable; or the inverter is limited to zero export
The tank reading is off by more than 5 percentage points, or above 100	The range and the tank's height as they were given (give the ranges again: a wrong height reads too high or too low); the transmitter at the bottom of the tank with its vented cable free; the transmitter on input 1
Pressure is off by more than 0.2 bar	The top of the range as it was given; the transmitter on input 2; check the channel with the loop calibrator
Flow is off by more than 10 percent	The top of the range as it was given (in litres a minute); the flow meter on input 3; its straight run in the pipe
The pump shows running when it is stopped (and the no-flow alert opens)	The pump's running contact is inverted: the electrician swaps its normally-open and normally-closed wires, or uses the opposite contact
The reading at 4, 12 and 20 mA is not the bottom, the middle and the top	Read the module from the box with `greecon-edge modbus-test` (unit 3, input registers 0 to 2; the command is in the hardware reference): 4, 12 and 20 mA must read 4000, 12000 and 20000. If they do, give the range again. If they do not, the loop is wrong (a wire, the 24 V supply, the wrong input): the electrician checks it
A test message did not arrive	The address, the number's country code, spam; look at **What was sent** under **Who is told**
After a power cut the box is not back within 3 minutes	The BIOS "power on after power loss" setting; the clock; `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`
After the internet cut the queue does not empty	`greecon-edge status` and the box's log say what it is waiting for; then the box is offline: see the runbook for an offline box

Write down what was wrong and what you changed, on the commissioning record. A reading that was wrong stays as it was, with a note: do not edit history.

Quick runbooks: the ones the project manager uses

For: the founder and the project manager. Each runbook says what you see, what to check, what to do, what to tell the customer, and what to write down afterwards. Commands are in `code`; places in the app are in **bold**.

The product exists so that people hear about problems before their guests do. These pages exist so that **Greecon hears about its own problems before its customers do.**

R0 The routine

When	What	How long
Every morning (the project manager; the founder when away)	Read the morning email from Greecon ("Greecon, Mon 19 Oct: …"). Open **Today**: the sites under **Needs a look**, worst first. For anything not **Stable**, follow the runbook below. If everything is **Stable**, do nothing.	5 min
Same day	A customer who is **Attention needed** or **Unknown**: phone them *before* they phone you ("we saw your box was offline since 03:10; we are looking into it").	
Every Friday (founder)	The pipeline sheet (the Sales kit (document 04, not part of these guides)); the hardware register; take the weekly backup (R13); look at Railway **Usage**	30 min
Every month (founder)	Restore drill (R13); read the last month's alerts at each live site with the owner: *useful? too many?* (R5); look at the audit log for anything unexpected; check Resend and Twilio balances; update packages (R14 lists who can do what)	2 hours
Every six months	Re-check the legal list (document 22); read the claims register (document 21) against what is being said; renew `security.txt` before 30 September 2027	half a day

Who owns what. The **founder** owns every incident about the platform or about security, and is the person the outside uptime check calls. The **project manager** is the first person for a customer's site and deputises for the founder. **Nobody promises a 24-hour response yet** (the customer promise is: acknowledged within four business hours, Monday to Saturday). The platform does the night shift: Act now reaches the *customer's own* people at once.

Severity.

	What	Target
S1	The platform is down; sign-in does not work for anyone; a possible leak of data	Acknowledge to affected customers within 30 minutes of finding out (a target, not a promise)
S2	One customer's site is silent or wrong (box offline, readings stale)	Same working day
S3	A wrong value, a message not received, a question	Within four business hours

While a clock runs. Safety events, security incidents and personal-data incidents have legal clocks (4 hours, 24 hours, 72 hours) that depend on the law that applies to the *customer*. R12 says what to do, and the clocks are in document 21, section 4. Write down the **time you found out** first.

R1 A customer's box is offline

You see: Today, under *Needs a look*, says *The box has not reported for N: it is offline*; or the customer's morning message says readings are out of date. **Or you are told:** Greecon's own team gets a message when a customer's box has been silent for a quarter of an hour (*A customer's box went silent*, also under **Who is told → What was sent**): the client, the site, since when, its last reading, and whether any message has gone to the customer (it has not, by this one). **By day it goes at once; in the team's quiet hours (22:00 to 07:00 unless changed under Who is told) it waits, shown there as *Waiting* with the time it is due, and goes when they end, unless the box has reported again by then, in which case it is dropped and the log says why:** a box that goes quiet at midnight is on **Today** at once and its message arrives at 07:00; nobody is woken for it. If three or more boxes go silent within ten minutes it is **one** message that says so: that is Greecon's platform or a provider they share, not that many customers losing power at once: go to R7. If the email service cannot take the message (it is down, or the day's allowance is used up) it is tried again every five minutes for six hours; after that, the morning message and **Today** are what say it. A box that keeps coming back and going again is told of three times a day at most, so a poor connection at one farm cannot fill the inboxes (it stays on **Today** while it is silent).

Check, in this order:

1. **Is the whole site offline?** Phone the owner or manager. *"Is the power on in the plant room? Is the router's light on?"* A power cut at the site is the commonest cause, and ends by itself.
2. **Is it the site's internet?** If they have other internet things down, it is the router or the line. Ask them to restart the router. (The box keeps about three days of readings, for a site of up to about thirteen devices, and sends them when the internet returns. **Nothing is lost** if it returns within that time.)

3. **Is it the box?** If power and internet are fine: on the box (keyboard and screen, or SSH on the site's network): `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`. Read the last lines: they say what is wrong (cannot reach Greecon, a clock far out, the credential refused).
4. **Is it Greecon?** If every customer's box is offline at once, the platform is down: R7.

Fix: restart the service `sudo systemctl restart greecon-edge`; if the clock is out `sudo timedatectl set-ntp true`; if the box's log says *the platform does not recognise this box's credential*, make a new code: on the site's setup page, **Install a box, or replace this one**, then run the new line on the box. A box that has been off for hours reports in again by itself.

Tell the customer: what you saw and when, what you did, what they need to do. If nothing was lost, say so.

Afterwards: a line in the incident log (date, site, cause, minutes offline). If it is the *same site twice*, find the cause: a loose plug, a router that restarts nightly.

R2 The box is online but a site has no readings, or a device "does not answer"

You see: the box is **Online** and the site is **Attention needed** or **Unknown**; the setup page shows the device as **Needs attention** with *no answer from 192.168.1.50:502 (unit 2)*, and on the box `greecon-edge status` lists it as *NOT OK*.

Check:

1. **Is the converter powered and on the network?** From the box: `ping 192.168.1.50` (its address on the setup page).
2. **Has the converter's address changed?** Press **Find equipment**. If the converter answers at a new address, the devices say **Found** there: R17.
3. **Does one device fail or all of them?** All: the converter or the cable. One: that device (its number, its power, its wires).
4. **From the box:** `greecon-edge modbus-test --host <address> --unit <n> --type input --address 52 --data-type float32`. *Could not read: no answer* means the device or the cable; *has no register at that address* means the register is wrong (the device answers); a value means the path works.
5. **A meter powered from the grid** goes dark in a power cut: expected, and it returns with the power.

Fix: reseat the converter's power; check **A** and **B** are not swapped; check the number on the device's label against the install card; check speed and parity match the converter's. The numbers and speeds are set at the office and are not changed on site. If a device was replaced, the new one is **numbered and set at the office first**, like the one it replaces (a new meter or module is delivered as number 1): fit the spare that was numbered there, then press **Find equipment**.

Tell the customer: which device, since when, and whether their alerts for it can be trusted meanwhile (they cannot while it does not answer: the platform shows *Unknown*, never *fine*).

R3 A reading looks wrong

You see: a value that is zero, never changes, is ten or a thousand times too big or small, is negative where it cannot be, or has import and export swapped.

Symptom	Likely cause	Fix
Zero all the time	A clamp on the wrong cable, a closed breaker upstream, the wrong register, or the inverter is limited to zero export	`modbus-test` against the meter's display; confirm the clamp is on the right phase
Constant	A frozen device, a wrong register kind (input vs holding)	`modbus-test` twice a minute apart
x1000 too big	Watts shown as kilowatts, on a reading mapped by hand (the Eastron meters already scale it)	In the register form: multiply by **0.001**
Tank above 100 percent, or a value that "cannot be a tank level"	The range or the tank's height was given wrongly, a wire has come off (the reading is below zero), or the transmitter is on the wrong input	Give the ranges again (this manual, the ranges step) and see R18
Import and export swapped	Clamps the wrong way round	The electrician turns them; repeat the direction check. Greecon notices on its own at night or in cloud (a Check today alert, *The grid meter may be fitted the wrong way round*, after five minutes of energy going out of a plant that nothing is producing for); at midday it cannot tell
The pump shows running when stopped	The contact is inverted	Swap the contact's normally-open and normally-closed wires, or use the opposite contact. Until it is right, and if a flow meter is fitted, the platform sees a running pump with no flow and opens the no-flow alert (an Act now) whenever the pump is stopped; when the pump really runs it shows stopped

Tell the customer when a number they relied on was wrong, and for how long. **Do not edit history:** a wrong reading stays as it was, with a note. The platform keeps what was measured.

R4 A customer did not get a message, or got one that was wrong

Check:

1. **Who is told** at that customer (sign in, **Open for setup**): is the person on the list, with the right email and number *with the country code*?

2. **What was sent**, on the same page: each message, to whom, when, and its result: **Sent** (the email or text provider took it), **Not sent** (the **Why** column says why), **Waiting**, or **Not needed** (for example, someone acknowledged the alert before it went). **Sent** does not mean the person saw it. After **Send a test message** the page says *Accepted for delivery*, which means the same, so ask the person whether it arrived.
3. **Email in spam?** Ask them to look, and to mark it not spam. **Text:** which network? (some operators replace a sender name; see the Founder Go-Live Guide (PDF 01), step 8).
4. **Quiet hours** hold Check today until morning; only Act now is sent at night.
5. **Hourly limits:** everything that is not Act now (Check today, and the notes that an alert was acknowledged or has cleared) is limited to 60 emails and 20 texts an hour per account; past that a message is dropped and the **Why** column says so. Act now has a limit of its own, 120 emails and 60 texts an hour, and past it a message is **held**, not dropped, and sent as soon as the hour allows. A condition flapping all day should not reach either; if it did, find out why.
6. **Is the platform's messaging off?** The API's log at start-up says so ("Alert messages are OFF").
7. **A message that could not be delivered** (a key refused, an empty text balance, the provider down): an alert message is tried again every five minutes for the first hour and every half hour after that, for as long as the alert is open and nobody has acknowledged it, and **Who is told** says "N alert messages could not be delivered, the last because …". Each failed try stays in the log as a line of its own. A Check today email that falls due in the quiet hours, for any reason, waits for the morning; Act now never waits. A way of sending that is not set up on the platform at all (no text provider yet, say) is shown in the log as **Not sent**, is tried once an hour rather than every five minutes, and is **not** counted as "could not be delivered": **Who is told** says once that text messages are off, and Greecon's morning message does not name the customer for it. Mend the cause (the key, the balance, the address); the next try sends it. Greecon's own morning message names the customers whose alert messages are refused or not arriving.
8. **A customer who is Suspended** is sent nothing at all, and what had been planned for it is dropped with the reason in the log. An alert that opened while the customer was suspended is messaged normally, from the next round after it is made active again. **An alert that was already part way through its messages when the customer was suspended is not told again from where it stood**: look at **Alerts** for that customer when you turn it back on, and tell the people yourself if something is still open.
9. **The link in an alert message** (*I have seen this*): each alert message holds a link that works for 72 hours and acknowledges that one alert, for the person it was sent to, without signing in. Opening it changes nothing; pressing its button does. It shows in the customer's **Audit log** as *Alert acknowledged with the link in an alert message sent to [name]*. A page that says *This link no longer works* means the 72 hours are over, or the alert has gone, or the account is suspended, or `JWT_SECRET` was changed (changing it ends every link, as it ends every session). Nothing is lost: the person acknowledges in Greecon instead.

Fix: correct the list and send a test message to the person, and confirm it arrived.

An Act now alert that nobody acknowledged (the messages went on down the list until it ended): Act now is passed down the list, the first person at once, the second 30 minutes after the alert began, the third 60 minutes after, and so on, until the list ends or someone says they have seen it. Two ways: the *I have seen this* link in the message, then its button (no sign-in; it works for 72 hours; see point 9), or **Acknowledge** on the **Alerts** page, which only a person who can sign in as an **Owner**, an **Administrator** or an **Operator** can press (a **Viewer** or an **Auditor** cannot). Replying to the email or the text, or phoning someone, does not stop it: the platform cannot hear that. So **tell everyone on the Who is told list about the link**, and give the people who should also look at the screens a login of their own with the Operator role: the customer's owner adds it under **Settings → People**, **Add a person** (a Greecon support session cannot add people). **Settings → People** shows who has a login and with which role.

R5 Too many alerts

An alert nobody acts on trains people to ignore the next. The aim in steady state is **one alert or fewer per site per day**, every one with an action.

1. List the alerts of the last two weeks with the owner. For each: *what did you do?* An alert with no action is a limit that is wrong or a reading that is noisy.
2. Change the **limits** (the tank's *Low below*, *Critical below*) to numbers the owner would act on. Do **not** mute an alert type. Fix its cause.
3. A reading that flaps around a limit: raise the gap between *Low* and *Critical*; the platform already raises one alert per condition and re-opens it within 15 minutes rather than raising a new one.
4. Write down what changed, so the next person knows why.

R6 A person cannot sign in, or someone has left

- **Forgot the password:** the account's **owner** opens **Settings** → **People** and chooses **Reset password** for that person (a temporary password is shown once; they choose their own at first sign-in).
- **A customer's owner is locked out and has no other administrator:** they phone Greecon. **Confirm by phone that it is them** (ask something only the owner would know: the site's name and address, the last alert, who else is on the list). The founder opens **Clients**, finds the client, opens **People** in the client's row, and presses **Reset password** for that person. A temporary password is shown once; give it to them by phone, not by email. It ends every session the person had open, and the reset is written in the customer's own audit trail under the staff member's name. Only the owner of Greecon's account can do it; a project manager can see the people and is told who to ask.
- **Wrong password too often:** sign-in is rate limited per account. Wait a few minutes.
- **A team member leaves:** the founder opens **Team** → **Turn off**. Their session ends on their next click, and so does any customer account they had open. Then **change** `JWT_SECRET` if they knew anything secret (the Founder Go-Live Guide (PDF 01)).
- **A customer's person leaves:** the customer's owner opens **Settings** → **People** and chooses **Turn off** for the person; it ends their session on their next click.
- **The founder is locked out:** R14.

R10 A box is replaced

A failed box, a stolen one, or a customer who moved the equipment.

1. On the site's setup page press **Install a box, or replace this one**. **Confirm.** A new line is shown, once.
2. Install the operating system on the new box, paste the line. The new box takes the old one's place; **the old one's credential stops working at once.**
3. Nothing else is entered again: addresses and registers are the site's, not the box's.
4. Write the change in the hardware register.

R11 A customer asks for their data, or to leave, or to be deleted

- **A copy of their data:** they can take it themselves, whenever they like, from **Reports**: every reading for the last 31 days, hourly averages for up to a year, their alerts, their audit log and how their account is set up. Every download is recorded in their audit trail. If they cannot, do it for them from inside their account (**Open for setup**; the download is then recorded under your name).
- **Every reading Greecon holds** (when they stop, or when they ask for more than **Reports** gives): **Reports** stops at 31 days of every reading and a year of hourly averages, so make the file by hand. On your laptop, as in R13, open `psql "$DATABASE_PUBLIC_URL"` and paste this as **one line** (`<client id>` is on **Clients**; `<client>` is a short name for the file):

  ```
  \copy (SELECT timestamp_utc, site_id, device_id, point_id, canonical_name, value_numeric,
  value_bool, value_text, unit, quality FROM telemetry_readings WHERE tenant_id = '<client id>'
  ORDER BY timestamp_utc, canonical_name, point_id) TO 'readings-<client>.csv' WITH (FORMAT
  csv, HEADER)
  ```

  The times are UTC. In that file the site, the device and the reading are only ids, so make the key to them as well and send both (also one line):

  ```
  \copy (SELECT s.name AS site, s.id AS site_id, d.name AS device, d.id AS device_id, p.label
  AS reading, p.canonical_name, p.id AS point_id FROM points p JOIN devices d ON d.id =
  p.device_id JOIN sites s ON s.id = p.site_id WHERE p.tenant_id = '<client id>' ORDER BY
  s.name, d.name, p.label) TO 'names-<client>.csv' WITH (FORMAT csv, HEADER)
  ```

  Greecon keeps one reading a minute for each reading point, so a hotel's ten points make about a million rows in 90 days: a large file, which a spreadsheet may not open whole.
- **The messages that were sent** (**What was sent**, under **Who is told**) are in none of the downloads. If the customer asks for them as a file, in the same way:

  ```
  \copy (SELECT created_at, due_at, sent_at, contact_name, channel, destination, kind,
  priority, subject, body, status, error FROM notifications WHERE tenant_id = '<client id>'
  ORDER BY created_at) TO 'messages-<client>.csv' WITH (FORMAT csv, HEADER)
  ```

- **Sending a file you made by hand:** it holds readings of a named site, and the message file holds the names, email addresses and phone numbers of the people who were told, so **not by plain email**. Put it in a shared-drive folder that only the customer's owner can open (by their email address, not "anyone with the link"), tell them by phone that it is there, and delete it when they confirm in writing that they have it. A download from **Reports** is written in the customer's audit trail; **a file made by hand is written in no audit trail**, so write it in the same sheet as the data-deletion log below: who asked, when, which files, how they were sent, when it was confirmed.
- **They stop at day 60 or day 90:** take the equipment away within 14 days; give them the files (the downloads from **Reports**, and every reading Greecon holds, made by hand as above); set the client **Suspended** (**Clients → More → Suspend this client**, which asks once and says what it will do) (it can be turned back on; if you do, read R4, point 8, about alerts that are still open); keep the account until they confirm in writing that they have their data.

- **Deletion (they ask in writing, after they have their data):** this is the only time anything is deleted. The founder does it, in a transaction, after the files are with the customer and the request is in writing:

```
-- in the Railway Postgres service, Data tab or psql with the public connection string
BEGIN;
DELETE FROM tenants WHERE id = '<client id, from Clients>';
DELETE FROM access_requests WHERE lower(email) = lower('<owner email>');
-- the first must say DELETE 1 (if not, ROLLBACK instead); the second says DELETE 1 if the
owner
-- asked through the request-access page from that address, and DELETE 0 if not. Then:
COMMIT;
```

The first statement removes the account and everything under it: sites, devices, readings, alerts, people, messages sent, audit trail. The database does this by itself: every table that holds a client's data is tied to the client's row, so it goes when the row does (the automatic test checks the people, the settings and the messages sent; for the other tables it follows from how they are built). The one table that is not tied to a client is the list of requests from the request-access page, which holds the name, company, email and phone the owner gave: that is what the second statement removes (if they asked from another address, run it again with that address). **Then restart the API service** (a redeploy) so that the platform's memory forgets them. **Write in the data-deletion log** (a sheet, outside the platform): who asked, when, what was deleted, when it was confirmed to them. **What stays, and what to tell the customer:** copies in **backups** disappear when the backups expire (Railway's daily ones within a week; the weekly dumps you keep: delete those that hold them as you rotate); Greecon's own morning messages to its team, already sent, still name the client and its sites in their text; Resend and Twilio keep their own records of the emails and texts they delivered, on their own schedule; and files already downloaded (by the customer, or by hand) stay where they are.

R15 Find equipment cannot find a device

You see: on the setup page, in the step **Where each device is on the network**, after **Find equipment**: a device marked **Not found** (*Nothing that looks like an Eastron meter answers at unit 2. Check the address set on it, the speed, and the A and B wires.*), or *No Modbus converter answered on this network*, or *A converter answers at 192.168.1.50, but nothing answers behind it*, or a device marked **Not sure which**.

Check, in this order:

1. **Is the box online?** Until it is, the button is grey, or the page says *The box did not pick this up*. Do that first (R1).

2. **Which sentence is it?**
   - *No Modbus converter answered on this network.* The converter is off, or not on the same network as the box (a guest network, a second router, a cable in the wrong socket). Look at its power and network lights, and check that the box and the converter are on the same router. At a desk, if the converter is on a different part of the network, or if it uses another port than 502, open **I know the converter's address**, type its address (each part without a leading zero: 192.168.1.50) and, if it is not 502, its port, and press **Look at this address**. The box does not look at itself (127.0.0.1).
   - *A converter answers …, but nothing answers behind it.* The network side works and the cable side does not: **A** and **B** swapped (swap the two wires at the converter); the converter not in **Modbus TCP to RTU (gateway)** mode or not at 9600 baud, 8 data bits, no parity, 1 stop bit; devices without power; a cable that is broken or branched.
   - **Not found** for one device while the others are found: that device's power (a meter powered from the grid is dark in a power cut), the number on its label against the install card, its **A** and **B** wires and the cable to it.
   - **Not sure which**: Greecon refuses to guess, because a wrong guess could swap the grid meter and the solar meter, and every figure with it. *Something answers at 192.168.1.60, unit 1, but not clearly enough to tell whether it is the three-phase SDM630 planned.* Press **Look again**, then check the speed and the A and B wires. *Units 1, 2 at 192.168.1.50 all answered with exactly the same words*: the converter is not in **Modbus TCP to RTU (gateway)** mode, so check its settings page against its label. Two of one kind at the same number on different converters, or at numbers that are not the kit's: read the label on each device, then use **Enter the address by hand** under each, with the number on its label. Two devices on one converter with the same number answer over each other: one was not numbered at the office, and numbers and speeds are not changed on site, so that part goes back there. Then press **Look again**.
3. **From the box:** `greecon-edge find` looks at the network as **Find equipment** does and prints what answers. Then read one device, with the unit number of the device you are looking for:

   ```
   greecon-edge find --host 192.168.1.50 --port 502
   greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 0
   greecon-edge modbus-test --host 192.168.1.50 --unit 5 --type discrete --address 0
   greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
   float32
   ```

   The first looks at one address; the others read the 4 to 20 mA module, the pump contact module and a meter. *Could not read: no answer* means the device or the cable; a value means the path works.
4. **A device the site does not have** is not a fault: press **The site does not have this: remove it** under it. **A real Waveshare module that is found but reads something odd:** its registers have not been tried on real hardware: write down what `find` and `modbus-test` print, and tell Greecon.

Fix: what the check found; then **Look again**. The step turns **Done** when every device has the address it was found at.

Tell the customer: only if a visit or a part is needed. Greecon only reads, so nothing of theirs is affected.

Afterwards: write down what was wrong and what you changed (the commissioning record, or the incident log if the site is live).

R16 A device was found at another address

You see: a device marked **Found**, and a button **Use this address** (or *Use these N addresses*): *Found at 192.168.1.60, unit 1: an Eastron meter, reading 1.80 kW at 231 V.* Nothing changes until you press it. Greecon fills in without you only a device that has no address yet, found at the kit's own number, as the model planned.

Check, in this order:

1. **Did the converter move?** Several devices **Found** at one new address, and the old address was the converter's: R17.
2. **Is the number the kit's?** *It is at unit 6, not 3: if it is this site's, use this address, and the install card gives 3 so that the next site looks like this one.* If it is this site's part, it works as it is: write it down, and do not change it on site. If it could be the customer's own device on the same cable, do not press the button: read its label and ask.
3. **Is it another model?** *It is a single-phase SDM230, not the three-phase SDM630 planned. If you use this address, Greecon reads it as a single-phase SDM230.* That is right where the supply is single-phase and the install card lists the meter fitted; if a three-phase meter was meant, do not press the button: the electrician changes the meter. The other way round (*It also answers the three-phase meter's total-power register, so it may be a three-phase SDM630*, on a **Found** or a **Set** line) the meter is read as an SDM230 and shows one phase only: check its label and, if it is an SDM630, remove this device and add the three-phase meter from **Add equipment Greecon already knows**.
4. **Is it something nobody planned?** It is listed under *Also answering, and not part of this site's equipment*: not yours, leave it alone.

Fix: if the line makes sense, press **Use this address**: Greecon sets the address (and reads a meter of the other model as that model), the box picks it up within a minute, and the device shows **Set**. The change is written in the audit trail, old and new. If the line does not make sense, do not press it: find out why first (R15).

Tell the customer: nothing, unless the cause is on their network (a router that changes the converter's address): ask for a fixed address (R17).

Afterwards: a line on the commissioning record: which device, which address, and why.

R17 The converter's address changed

You see: the readings of every device behind one converter stop together (*Last reading N ago. It has stopped arriving*), and each shows **Needs attention**: *no answer from 192.168.1.50:502 (unit 1)*. The box is **Online** and the converter's lights are on. The customer may say that the router was restarted: a router hands out addresses, and it may give the converter a new one.

Check: the box is online, the converter is powered, and more than one device behind it fails at the same moment. From the box, `ping 192.168.1.50` (the address on the setup page) gets no answer, while `greecon-edge find` shows a converter at another address.

Fix: press **Find equipment**. The box finds the converter at its new address and each device behind it says **Found** there. Press **Use these N addresses**: the devices turn **Set** and the box picks them up within a minute. The readings return; the gap stays a gap, because the box could not read the devices while it did not know where they were. If nothing is found, the converter is off or on another network: R15.

To stop it happening again: ask the customer's network administrator for a fixed address for the converter (a "DHCP reservation" on its MAC address in the router): the second message on **The install card** asks for it. Write the address on the converter's label and the handover record.

Tell the customer: what you saw and when, that nothing of theirs was affected (Greecon only reads), and the request for a fixed address.

Afterwards: a line in the incident log (date, site, minutes without readings).

R18 A reading is impossible after the ranges were given

You see: in the step **Readings arrive and look right**, a transmitter's reading cannot be real: *-31.2 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.*, or *133 cannot be a tank level (0 to 100)*, and likewise for the line pressure and the flow. Or it is believable, but disagrees with a dipstick or a gauge.

Check, in this order:

1. **Below zero: a wire off.** The 4 to 20 mA module is left in its 0 to 20 mA mode on purpose, so that a wire that has come off, or a loop that has lost its 24 V supply, reads **0**: below the 4 mA of an empty tank, and so below zero once the range is applied (about -31 for a 0 to 5 m level transmitter in a tank 4 m tall). Read the module from the box:

   ```
   greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 0
   ```

   Address 0 is input 1, 1 is input 2, 2 is input 3. **0 means no current.** A working transmitter reads at least 4000 (4 mA, counted in microamps). Is it on the input the install card says (level 1, pressure 2, flow 3)? The electrician checks the loop and its 24 V supply, with the lock-out where needed: you do not work on it.
2. **Far too high or too low: the range or the tank's height was given wrongly.** Read the label again. Too high: the range given is bigger, or the height smaller, than the real one (a tank 4 m tall given as 3 m, with a 0 to 5 m transmitter, reads 133 when it is full). Too low: the other way (a label in bar typed as metres reads about ten times too low: choose bar under **Unit on the label**). Measure the tank inside again, from the bottom, where the transmitter hangs, to the top of the water when it is full. Under **Which register holds which reading** give the ranges again and press **Save the ranges**; the form does not show the level's answer back, so type both numbers. It then says what it set, for example *Set now: 4 mA is 0 % and 20 mA is 125 %.*
3. **A transmitter shorter than the tank** is refused when you save (*it cannot see the top of the water*): fit one with a longer range.
4. **A loop calibrator** in place of the transmitter settles it: send 4, 12 and 20 mA, and the module must read 4000, 12000 and 20000. If it does and Greecon is still wrong, it is the range or the height; if it does not, the wiring, the module's input or the loop.
5. **A transmitter the site does not have** (no flow meter) reads 0 and looks like a wire off: tick **Not fitted at this site** under it.

Fix: what the check found. A reading that was wrong stays as it was, with a note: do not edit history.

Tell the customer when a number they relied on was wrong, and for how long.

Afterwards: write down what was wrong (a wire, a range, a height) and what you changed.

APPENDICES

Forms, words, symptoms, shopping lists and contacts

These pages are for printing and for looking things up. Appendix A has the blank forms for any site (each site pack has the same forms made for its kit). Appendix B explains the words. Appendix C is a way in by symptom. Appendix D has the shopping lists with room for three quotes. Appendix E is the contact sheet and a card of the numbers that never change.

A Forms to print

Print the form, take it to the site, and file it in the customer's folder afterwards. The forms are general: cross out what the site does not have. Each form starts on a new page.

- ☐ A.1 Survey form
- ☐ A.2 Install-day checklist
- ☐ A.3 Commissioning record
- ☐ A.4 Handover record
- ☐ A.5 Rehearsal record

A.1 Survey form

ANY SITE

Field	Answer
Customer, site name and address, date	
Contact: name, mobile, email (first, second, third to be told)	
What went wrong last year (their words)	
Template	☐ Hotel or guesthouse ☐ Solar plant with grid connection ☐ Farm or greenhouse with irrigation ☐ Water tank and pump station
Prices from the bills: electricity per kWh; paid for energy sent to the grid; water per m^3	
Twelve electricity bills and the water bills collected	☐ Yes ☐ Later
Supply: single or three phase; main breaker size (A)	
Inverter: brand, size (kW), AC output cable location; allowed to send to the grid?	
Tank: height (m), water source; pump (kW); starter location; pipe pressure (bar)	
Meters needed (grid, solar, load, pump) and where	
Sensors needed (level, pressure, flow, pump running; soil, weather) and where	
Network: customer's cable or cellular; router brand; who can give an address	
Power for the box: socket location	
Electrician: name, mobile, day	
Photographs taken (list)	
Go / no-go, and why	☐ Go ☐ No-go

A.2 Install-day checklist

ANY SITE

The electrician and the project manager work at the same time, in this order. Cross out the rows that do not apply to the kit.

	#	Who	Do this
☐	1	ELECTRICIAN / PROJECT MANAGER	Before anyone touches anything: the signed order form and the data terms are in your hand, the electrician is there, and the lock-out is agreed. Walk the site together. Agree where each meter and sensor goes. Photograph everything before you start.
☐	2	ELECTRICIAN	Apply the customer's lock-out on the board to be worked on.
☐	3	ELECTRICIAN	Fit the meters (grid, solar, load or pump): voltage taps with their fuses or small breakers, current clamps with the arrow towards the site.
☐	4	ELECTRICIAN	Connect the pump's run contact (a spare auxiliary contact, as a dry contact) to input 1 of the pump contact module. Never to a control circuit, and never to the module's relay outputs.
☐	5	ELECTRICIAN	Fit the level, pressure and flow transmitters, with the customer's plumber for the wet parts, and wire their 4 to 20 mA loops with the 24 V supply in series, as the module's manual draws it: level on input 1, pressure on input 2, flow meter on input 3.
☐	6	PROJECT MANAGER	Mount the enclosure, the converter, the two modules, the 24 V supply, the UPS and the box.
☐	7	PROJECT MANAGER	Run the RS-485 data cable: one cable, a daisy chain, no branches. The electrician connects its ends at the meters inside the board.
☐	8	PROJECT MANAGER	Check the number on each device's label against the install card (grid meter 1, solar meter 2, 4 to 20 mA module 3, load or pump meter 4, pump contact module 5, soil probe 6, weather sensor 7). They were set at the office: do not change them. Photograph each label.
☐	9	PROJECT MANAGER	Check the converter's label: Modbus TCP to RTU gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502. Ask for a fixed address for it and write the address down.
☐	10	PROJECT MANAGER	Connect the network: the customer's router, or the 4G router with its SIM.
☐	11	ELECTRICIAN / PROJECT MANAGER	Photograph before closing: every meter's label, the wiring, the converter's settings. Close the board.
☐	12	ELECTRICIAN	Remove the lock-out only when you both agree the work is closed up.
☐	13	PROJECT MANAGER	Power up: UPS, supply, converter, box.
☐	14	PROJECT MANAGER	Open the customer for setup and go down the setup list: press **Find equipment**, give the ranges, and write down the electrician's name and licence number.
☐	15	ELECTRICIAN	Stay until commissioning is done.

A.3 Commissioning record

ANY SITE

Compare in this order and write the pair of numbers. Do not start, stop or open equipment yourself: the owner's own person does.

Check and how	Pass when	Greecon shows	Real	Result
Every meter's display against Greecon. Read both at the same time.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail
Grid direction. A known load on, the sun down.	Import rises, export does not			☐ Pass ☐ Fail
Solar at night.	Zero, or a few watts			☐ Pass ☐ Fail
Solar at midday. Against the inverter's display.	Within about 5 percent			☐ Pass ☐ Fail
Tank level. A dipstick or the sight gauge, at two levels.	Within 5 percentage points			☐ Pass ☐ Fail
Pressure. Against the gauge on the same line.	Within 0.2 bar			☐ Pass ☐ Fail
Flow. A bucket test, or the flow meter's display.	Within 10 percent			☐ Pass ☐ Fail
Pump running. The owner starts the pump.	Shown within 30 seconds, changes both ways			☐ Pass ☐ Fail
Soil probe. In the air, in a glass of water, in the soil.	Three different readings			☐ Pass ☐ Fail
4 to 20 mA channels. With a loop calibrator, send 4, 12 and 20 mA.	Bottom, middle and top			☐ Pass ☐ Fail
A test message. Each person on the list.	It arrived, in the inbox and on the phone			☐ Pass ☐ Fail
Power cut. Unplug the box for 30 seconds, plug it in.	Back reporting within 3 minutes, and the box says nothing about readings lost			☐ Pass ☐ Fail
Internet cut (the first install, and any time the box or its software is new). See the box above.	A queue while it is out; empty a few minutes after it returns			☐ Pass ☐ Fail

Signed off by: _____ Date: _____ Site: _____

A.4 Handover record

ANY SITE

Field	Answer
Customer and site	
Date, and who was there	
Who is told, in order (name, mobile with country code, email)	
Alert limits agreed with the customer	
Papers: who signed for the customer, and the date	
Electrician: name and licence number (or: no electrical work was needed)	
Anything the site cannot raise (no flow meter, no pump contact), said to the owner	
Prices entered and where they came from	
Setup list: every step Done and the site Commissioned	☐ Yes
Handover sheet printed and signed	☐ Yes
Shown: Overview, Energy now or the site's page, Value, Who is told with a live test message, Reports	☐ Yes
Left: label on the box; card with `info@greecon.earth` and your phone	☐ Yes
Photographs and survey form filed in the customer's folder	☐ Yes

Project manager: name and signature

Owner: name and signature

A.5 Rehearsal record

ON A DESK

Once before the first real site, once more with the first real kit, and again when anything in the manual changes (section 3.11). An afternoon, a laptop, and the box. Date: _____ Done by: _____

	#	Who	Step
☐	1	PROJECT MANAGER	Run the pretend kit on the laptop: `node scripts/pretend-meter.mjs --kit --port 5020`. It answers as unit 1 grid meter, 2 solar meter, 3 4 to 20 mA module (tank 12 mA, pressure 8.96 mA, flow 5.12 mA) and 5 pump contact module (closed).
☐	2	PROJECT MANAGER	On **Clients** press **Onboard a client** for a client named Rehearsal, with a second address of your own for the owner. Open it for setup and make a site from the **Hotel or guesthouse** template. Write that the papers are signed (**The papers are signed**).
☐	3	PROJECT MANAGER	Add the box, paste the line on the box, and watch **Install the box** turn Done.
☐	4	PROJECT MANAGER	The pretend kit listens on port 5020, not 502, so the box does not find it by itself: open **I know the converter's address**, type the laptop's address on the network (not 127.0.0.1) and port 5020, and press **Look at this address**. The grid meter, the solar meter, the 4 to 20 mA module and the pump contact module are each **Set**.
☐	5	PROJECT MANAGER	Give the ranges: level transmitter 5 and tank 4, pressure transmitter 10, flow meter 200, then **Save the ranges**. Tank level reads 62.5 %, line pressure 3.1 bar, pump flow 14 litres a minute, and Pump running reads 1.
☐	6	PROJECT MANAGER	Watch every step turn Done. Add yourself under **Who is told**, send a test message, write "no electrical work was needed" for the electrician, and confirm the steps that need a person. Commission the rehearsal site.
☐	7	PROJECT MANAGER	Stop the pretend kit. After about five minutes **Today** says, under **Needs a look**, that the readings are out of date. Start it again, and raise the tank's limits above 62.5 (Low below 80, Critical below 70): the tank is critical, an **Act now**. It arrives by email at once and by text ten minutes later unless you press **Acknowledge** or open the *I have seen this* link in the message.
☐	8	PROJECT MANAGER	Print the handover sheet. Download the readings from **Reports** (**Downloads**) and open the file in a spreadsheet.
☐	9	PROJECT MANAGER	Clean up: remove the rehearsal site's equipment and then the site, or suspend the client, so that it does not appear in the morning check. Keep it if you will use it for a sales demonstration.
☐	10	PROJECT MANAGER	With the first real kit: do steps 1 to 6 again with the real converter and devices on the desk in place of the pretend kit, and write down anything that differs from this book.

What did not match the manual, and what I changed

B Words

B.1 Words used in the manual

Word	Meaning
Box	The small computer at the site that reads devices and reports to Greecon
Converter	Puts the RS-485 cable on the network (a Modbus gateway: Modbus TCP to RTU). Set at the office
Modbus	The language most meters and modules speak. A device has a **unit number** and **registers**
Register	A numbered place in a device that holds one value. **Input** registers are read-only, **holding** can be written (Greecon never writes)
Clamp / CT	A ring that goes around a cable and tells the meter how much current flows
4 to 20 mA	A signal wire: 4 mA means the bottom of the transmitter's range, 20 mA the top. The Waveshare module reads it in microamps: 4000 to 20000. A wire that has come off reads 0
Range (of a transmitter)	The two ends printed on a 4 to 20 mA transmitter's label, for example 0 to 5 m or 0 to 10 bar. Greecon asks for it, in the label's words, and works out the scale
Approved equipment	The models a kit is made of, and no others: the Eastron SDM630 and SDM230 meters, the Waveshare Modbus RTU Analog Input 8CH and the Waveshare Modbus RTU IO 8CH. Greecon knows their registers and settings
Find equipment	The button that makes the box look at the site's network and find where each device is. It fills in what is certain and asks you about the rest. It only reads
Install card	The page for each site with what to bring, what each device is set to, the converter's settings, the visit in order, and three messages for the customer
Pump contact module	The module that reads whether the pump is running, from a dry contact on its starter. It only reads
Commissioning	Checking that what the platform shows matches the real world, once, on site
Act now / Check today	The two kinds of alert that reach a person. Everything else is only shown

B.2 More words

Word	Meaning
Unit number	A number that tells the devices on one cable apart. Every device has its own, and in every kit it is the same: grid meter 1, solar meter 2, 4 to 20 mA module 3, load or pump meter 4, pump contact module 5, soil probe 6, weather sensor 7. They are set at the office; the meters and the two modules are delivered as 1
Gateway mode	The converter's mode (Modbus TCP to RTU) in which it turns the box's requests into requests on the data cable. Not transparent, not raw TCP, not RTU over TCP
Speed, parity	The serial settings of the data cable. Greecon's standard is 9600 baud, 8 data bits, no parity, 1 stop bit, the same on every device and on the converter. They are set at the office
RS-485, A and B	The two-wire data cable the meters and modules share. A and B swapped is the commonest mistake
Daisy chain	One cable from device to device, with no branches
Lock-out	The customer's own procedure for switching a board off and keeping it off while the electrician works in it
CT ratio	On a meter with 5 A current transformers, the ratio set in the meter's menu (for example 250 for 250/5 A) so that it shows the real power
Loop calibrator	A hand-held that sends 4, 12 and 20 mA, to check a 4 to 20 mA channel and the range given for it
Microamps	How the 4 to 20 mA module reports a signal: 4 mA reads 4000, 12 mA reads 12000, 20 mA reads 20000. A wire that has come off reads 0
Dry contact	A switch that only opens and closes and carries no voltage of its own, like the spare contact in a pump's starter. The pump contact module reads whether it is closed
Fixed address, reservation	A network address the router always gives the converter (a "DHCP reservation" on its MAC address), or one set in the converter itself
Heartbeat	The box reports every 30 seconds, so that Greecon knows it is alive
Queue	The readings the box keeps while the internet is down: about three days for a site of up to about thirteen devices
Template	The plan of a kind of site in the platform: the equipment it usually has, with the readings each gives
Setup list	The platform's own list of up to twelve steps for a site. A step that does not apply is left out
Handover sheet	One printable page per site: where everything is, who is told, who commissioned it and when
Install line, code	The line shown once on the setup page that installs the box. Its code works once, for seven days
Owner, Administrator, Operator, Viewer, Auditor	A person's role in the customer's account. An Owner, an Administrator or an Operator can press Acknowledge in the app; a Viewer or an Auditor cannot (anyone a message was sent to can use the link in it)
Morning message	The one email at 07:30 that says whether everything is all right
Quiet hours	22:00 to 07:00: everything is held except Act now

Word	Meaning
Zero-export	A plant that is not allowed to send energy to the grid. Tick the box in the prices
Pretend kit	A program for the laptop that answers like a whole kit (two meters, the 4 to 20 mA module and the pump contact module), for practice

C Symptoms

Start from what you see. Each table says what it probably is and what to do; the longer answers are in the runbooks.

C.1 On site and at the box

Short answers; the longer ones are in the runbooks.

What you see	Likely cause	What to do
The install line stops with `STOP:`	The sentence says: no internet (or a firewall, or a clock far out), code used or expired	Do what it says. Before step 5 run the same line again; after it, run the line with no code at the end (section 3.6)
The box says Offline	Power, network cable, or the site's internet is down	Is the box's light on? Is the router up? On the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`
Find equipment finds nothing, or a device says Not found	The converter is off, or on another network than the box; A and B swapped; the speed or the mode at the converter; a device with no power	Read the sentence on the screen. On the box, `greecon-edge find` says what answers. The runbooks, R15
A device says Found, or Not sure which	Found at another address than the one it has (the router gave the converter a new one), at a number that is not the kit's, or as the other model of meter; or two of a kind answer at one number, or every number answers alike	Read the line. Press **Use this address** only if it makes sense. Numbers are not changed on site. The runbooks, R16 and the runbooks
A device needs attention (on the box: NOT OK)	The converter's address changed, the number on the device's label, speed, A and B swapped, converter not in gateway mode	Press **Find equipment** again; then `greecon-edge modbus-test` against the device; check each in that order
A reading is zero or constant	A wrong register on equipment mapped by hand, the wrong register kind (input vs holding), a clamp on the wrong cable	`modbus-test` raw numbers against the manual's table
A reading is ten or a thousand times too big or small	A wrong scale (W against kW) on equipment mapped by hand, or a range or a tank's height given wrongly on a transmitter	By hand: re-enter the conversion. Transmitter: give the ranges again; check with the calibrator

What you see	Likely cause	What to do
A reading cannot be real (*cannot be a tank level*, below zero, or far above 100)	A wire that has come off (the module reads 0), the range or the tank's height given wrongly, a transmitter on the wrong input	The runbooks, R18
Import and export are swapped	The clamps are the wrong way round	Electrician turns them; repeat 3.8.9
The clock is out	The box has no time service	`timedatectl set-ntp true`; the setup page names it ("its clock is N minutes fast")
The test message did not arrive	A wrong address, spam, or the number without its country code	Check **What was sent** under **Who is told** (it shows what happened); try the other way
The test email arrived in spam	The sender is new to that mail provider	Mark it not spam; ask them to add `alerts@greecon.earth` to contacts
A reading went Unknown after a while	The meter or the converter lost power or the cable came loose, or the converter's address changed	The setup page says which device and why; press **Find equipment**

C.2 A reading looks wrong

A value that is zero, never changes, is ten or a thousand times too big or small, is negative where it cannot be, or has import and export swapped.

Symptom	Likely cause	Fix
Zero all the time	A clamp on the wrong cable, a closed breaker upstream, the wrong register, or the inverter is limited to zero export	`modbus-test` against the meter's display; confirm the clamp is on the right phase
Constant	A frozen device, a wrong register kind (input vs holding)	`modbus-test` twice a minute apart
x1000 too big	Watts shown as kilowatts, on a reading mapped by hand (the Eastron meters already scale it)	In the register form: multiply by **0.001**
Tank above 100 percent, or a value that "cannot be a tank level"	The range or the tank's height was given wrongly, a wire has come off (the reading is below zero), or the transmitter is on the wrong input	Give the ranges again (this manual, the ranges step) and see R18
Import and export swapped	Clamps the wrong way round	The electrician turns them; repeat the direction check. Greecon notices on its own at night or in cloud (a Check today alert, *The grid meter may be fitted the wrong way round*, after five minutes of energy going out of a plant that nothing is producing for); at midday it cannot tell
The pump shows running when stopped	The contact is inverted	Swap the contact's normally-open and normally-closed wires, or use the opposite contact. Until it is right, and if a flow meter is fitted, the platform sees a running pump with no flow and opens the no-flow alert (an Act now) whenever the pump is stopped; when the pump really runs it shows stopped

C.3 A device does not answer the box

The setup list shows the device as **Needs attention**, with *no answer from 192.168.1.50:502 (unit 2)*, and on the box `greecon-edge status` lists it as NOT OK. Look in this order:

1. **The converter's address has changed** (the router gave it a new one): press **Find equipment** again. The box looks for the converter, and each device behind it says **Found** at the new address: press **Use this address**.
2. The **number on the device's label** is not the one on the install card, or two devices carry the same number (Find equipment says **Not sure which**).
3. A wrong **speed or parity** at the converter: it must match every device (9600 baud, 8 data bits, no parity, 1 stop bit).
4. **A and B swapped** on the data cable: swap the two wires at the converter.

5. The converter **not in gateway mode** (Modbus TCP to RTU).
6. The device has **no power**: a meter powered from the grid goes dark in a power cut, and returns with the power.

Nothing answers at all: the converter is off, or it is on another network than the box. Some readings work and others do not: a register address that is wrong for that device (on a probe or a battery mapped by hand), or, on a Waveshare module, a table in Greecon that differs from the real module (no real module has been tried yet, so write down what the box says and report it to Greecon). Test one register from the box:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
float32
greecon-edge modbus-test --host 192.168.1.50 --unit 5 --type discrete --address 0
greecon-edge find
```

Could not read: no answer means the device or the cable. *Has no register at that address* means the register is wrong (the device answers). A value means the path works. `greecon-edge find` looks at the network and says what answers, as **Find equipment** does.

D Shopping lists

One list for each kit, with room for three quotes. Ask three suppliers for the quote on the specification, not the brand, and keep the quotes in the shared drive. The estimates are planning figures in EUR, excluding VAT and installation labour, from public shop prices seen in October 2026: they are not quotes. Buy and bench-test one kit before buying five. For each part keep the make and model, the supplier, the date, the serial number or the batch, and the invoice.

H Hotel or guesthouse

Planning total: About EUR 800 to 1,800; typically 1,100 to 1,300. Add the electrician's time (half a day to a day) and yours.

	Item and specification	Qty	Planning estimate (EUR)	Quote 1	Quote 2	Quote 3
☐	**Box**. Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300			
☐	**Converter**. RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)			
☐	**Three-phase meter**. Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers	2	90 to 180 each			
☐	**4 to 20 mA input module**. Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60			

	Item and specification	Qty	Planning estimate (EUR)	Quote 1	Quote 2	Quote 3
☐	**Pump contact module**. Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used	1	30 to 60			
☐	**Level transmitter**. Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400			
☐	**Pressure transmitter**. 4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150			
☐	**Flow meter** *(optional)*. 4 to 20 mA output. Needed for "the pump is running but no water is flowing"	0 or 1	150 to 500			
☐	**24 V DIN power supply**. 30 to 60 W	1	25 to 50			
☐	**UPS**. A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150			
☐	**Enclosure and fittings**. IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150			
☐	**Cable and labels**. Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80			
☐	**Cellular router** *(optional)*. 4G, DIN rail, data SIM	0 or 1	150 to 200			

S Solar plant with grid connection

Planning total: About EUR 600 to 1,200.

	Item and specification	Qty	Planning estimate (EUR)	Quote 1	Quote 2	Quote 3
☐	**Box**. Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300			
☐	**Converter**. RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)			
☐	**Three-phase meter**. Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers	2, or 3	90 to 180 each			
☐	**24 V DIN power supply**. 30 to 60 W	1	25 to 50			
☐	**UPS**. A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150			
☐	**Enclosure and fittings**. IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters’ voltage taps, glands	1	80 to 150			
☐	**Cable and labels**. Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80			
☐	**Cellular router** *(optional)*. 4G, DIN rail, data SIM	0 or 1	150 to 200			

F Farm or greenhouse with irrigation

Planning total: About EUR 700 to 1,500.

	Item and specification	Qty	Planning estimate (EUR)	Quote 1	Quote 2	Quote 3
☐	**Box**. Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300			
☐	**Converter**. RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)			
☐	**4 to 20 mA input module**. Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60			
☐	**Level transmitter**. Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400			
☐	**Pressure transmitter**. 4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150			
☐	**24 V DIN power supply**. 30 to 60 W	1	25 to 50			
☐	**Enclosure and fittings**. IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150			
☐	**Cable and labels**. Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80			
☐	**Cellular router**. 4G, DIN rail, data SIM	0 or 1	150 to 200			
☐	**Soil moisture probe**. RS-485 Modbus, with a published register table, one for each zone	1 for each zone	40 to 120 each			
☐	**Temperature and humidity sensor**. RS-485 Modbus, with a published register table	1	20 to 80			
☐	**Pump contact module** *(optional)*. Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used	0 or 1	30 to 60			
☐	**Flow meter** *(optional)*. 4 to 20 mA output. Needed for "the pump is running but no water is flowing"	0 or 1	150 to 500			

W Water tank and pump station

Planning total: About EUR 650 to 1,300.

	Item and specification	Qty	Planning estimate (EUR)	Quote 1	Quote 2	Quote 3
☐	**Box**. Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300			
☐	**Converter**. RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)			
☐	**4 to 20 mA input module**. Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60			
☐	**Pump contact module**. Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used	1	30 to 60			
☐	**Level transmitter**. Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400			
☐	**Pressure transmitter**. 4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150			
☐	**Flow meter** *(optional)*. 4 to 20 mA output. Needed for "the pump is running but no water is flowing"	0 or 1	150 to 500			
☐	**24 V DIN power supply**. 30 to 60 W	1	25 to 50			
☐	**UPS**. A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150			
☐	**Enclosure and fittings**. IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150			
☐	**Cable and labels**. Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80			
☐	**Cellular router** *(optional)*. 4G, DIN rail, data SIM	0 or 1	150 to 200			
☐	**Single-phase meter** *(optional)*. Eastron SDM230, on the pump's feed, to measure its electricity	0 or 1	35 to 60			

The spares kit and the tools

One spares kit for each project manager: one of each, so that a failed part is swapped on the day.

- [ ] A converter
- [ ] A 4 to 20 mA module
- [ ] A pump contact module
- [ ] A meter
- [ ] A 24 V supply
- [ ] A transmitter (pressure)
- [ ] A spare box, pre-installed, with no code, ready to take one
- [ ] Network cables
- [ ] A loop calibrator
- [ ] A multimeter
- [ ] Fuses
- [ ] A spare SIM

The project manager's tools:

- [ ] A laptop
- [ ] A USB-to-RS-485 adapter (to number a module or a probe, and to talk to one device alone)
- [ ] A loop calibrator (or a 4 to 20 mA source)
- [ ] A multimeter
- [ ] Screwdrivers (slotted and Phillips, small and medium)
- [ ] Wire strippers and a ferrule crimper
- [ ] A network cable tester
- [ ] Cable ties and labels
- [ ] A head torch
- [ ] Spare fuses and terminals
- [ ] A phone for photographs

E Contacts and the numbers that never change

E.1 Contacts

Fill in, print, and keep a copy in the password manager.

Who	Phone	Email
Founder		eridon.manuka@greecon.earth
Project manager		
Electrician (first call)		
Electrician (second)		
Railway support		station.railway.com
The lawyer		
The insurer		
Each customer's first person	(on its handover sheet)	

What	Address
The web app (customers, staff, the project manager)	`https://app.greecon.earth`
The platform's own address, which every box reports to. **Never change it once a box exists**	`https://api.greecon.earth`
Where alert emails come from	`alerts@greecon.earth` (replies go to `info@greecon.earth`)

E.2 The numbers that never change

What	Value
Unit numbers (set at the office)	Grid meter 1, solar meter 2, 4 to 20 mA water module 3, load meter or pump meter 4, pump contact module 5, soil probe 6, weather sensor 7. A battery is addressed by hand (for example 8)
The data cable and the converter	9600 baud, 8 data bits, no parity, 1 stop bit; converter on port 502, in Modbus TCP to RTU gateway mode, at a fixed address. Set at the office
Eastron meters	Total active power: input register 52 on the SDM630, 12 on the SDM230; 32-bit floating point, high half first; watts to kW by 0.001
A register counted from zero (equipment mapped by hand)	A manual's 30053 is address 52; 40001 is address 0
The 4 to 20 mA module	Inputs 1, 2 and 3: level, pressure, flow. In its default 0 to 20 mA mode it reads microamps: 4 mA is 4000 and 20 mA is 20000; a wire that has come off reads 0

What	Value
A tank level from a transmitter longer than the tank	Greecon asks for the range on the label (in metres or bar) and how tall the tank is, and shows a percentage of the tank. 20 mA reads range ÷ tank height × 100 (a 0 to 5 m transmitter in a 4 m tank: 125); a full tank reads 100
What Find equipment looks at	Private addresses on the site's own network, port 502 (or the one you give), unit numbers 1 to 10; reads only; about a minute at most
The box needs outside	Outgoing HTTPS (443) to `api.greecon.earth` (and `nodejs.org` at install) and a time service (UDP 123). Nothing needs to come in
The box keeps	About three days of readings, for a site of up to about thirteen devices. It reports every 30 seconds
The install code	Works once, for seven days
A setup session in a customer's account	Four hours
Test messages	Six an hour for the whole account
Standard limits	Tank Critical below 15 percent (Act now), Low below 35 percent (Check today); pump no flow at or below 0.2 litres a minute held for 30 seconds (Act now); battery Low below 25 percent
How alerts reach people	Act now: email at once, text after ten minutes if not acknowledged, the next person at 30 minutes. Check today: one email, held until morning at night. Morning message 07:30. Quiet hours 22:00 to 07:00 hold everything except Act now. Resolved after 15 minutes clear
After install	Day 1, 3 and 7 checks, then the morning check; day 30 call; day 60 decision meeting; day 90 convert or remove within 14 days

```
greecon-edge status
journalctl -u greecon-edge -n 50 --no-pager
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
float32
greecon-edge find
sudo systemctl restart greecon-edge
sudo timedatectl set-ntp true
```